Hourly Air-Quality Forecasting for Nashik Monitoring Stations with KNN Imputation and a CNN-LSTM Network: 2025 Backtest and Partial 2026 Evaluation

Academic Project Report

[Student name] · [Roll number]

[Department, Institution]

[Course / guide]

[Submission date]

Abstract

Hourly concentrations of seven pollutants (PM2.5, PM10, NO2, SO2, CO, O3 and NH3) from four continuous monitoring stations in Nashik, India (MIDC Ambad, Gangapur Road, Pandav Nagari and Hirwadi) were used to train one convolutional-LSTM (CNN-LSTM) network per station that predicts the next hour from the previous 24. Missing inputs were filled with a K-nearest-neighbours imputer fitted on training data only. The pipeline runs in two separate modes. The 2025 backtest uses data up to 31 December 2024 (network fitted to 30 June 2024, validated on July-December 2024) and is scored on 2025, which is measured at all stations. The 2026 run uses data up to 31 December 2025 in the same way and forecasts all 8,760 hours of 2026 at all four stations; measurements exist only for January-June and August 2026 at three stations and not at all at Pandav Nagari. Two kinds of prediction are scored separately on measured hours: a recursive forecast of the whole year from the forecast origin, which uses no later measurement, and a rolling one-step evaluation, in which the frozen model predicts each measured hour from the measured previous 24 hours. Faulty sensor readings (stuck values, CO zeros and near-constant 'flatline' days) are removed before training and are never scored. Pooled over stations, with quality-flagged sensor series excluded, the recursive forecast changed RMSE relative to a month-by-hour climatology by -223.0% (PM2.5) to -20.9% (PM10) in 2025 and -211.8% (PM10) to 10.6% (CO) in 2026 (negative = worse than climatology): fed its own predictions for a year, the network is a worse year-ahead estimate than climatology for every pollutant in 2025 and 6 of 7 pollutants in 2026. The one-step predictions changed RMSE relative to persistence by -3.7% (NO2) to 8.1% (SO2) in 2025 and -5.8% (CO) to 10.4% (PM10) in 2026: a small, mostly consistent gain when fresh measurements are available every hour. Daily CPCB AQI values were derived from the predicted concentrations; they are model estimates, not official AQI bulletins.

Keywords

air-quality forecasting; CNN-LSTM; KNN imputation; recursive forecasting; backtest; CPCB AQI; Nashik; chronological validation

Key Findings in Plain Language

This page states the conclusions first. The rest of the report gives the evidence.

- What the model does. For each station, a neural network reads the last 24 hours of seven pollutant measurements and predicts the next hour.
- Next-hour prediction works, modestly. The simplest possible guess is persistence: 'next hour = the latest measured value'. One hour ahead, the CNN-LSTM is more accurate than persistence for 22 of 28 station-pollutant series in 2025 and 12 of 17 in 2026. The typical (median) error reduction is 5.8% in 2025 and 4.4% in 2026. That is a real but small improvement.
- A high R^2 is not proof of a good model. Hourly pollution changes slowly, so persistence already reaches a high R^2. Every R^2 in this report is shown next to a baseline scored on exactly the same hours.
- A full-year forecast does not work with this model. Feeding the model its own predictions for a whole year lets errors pile up. Against climatology (the average for that month and hour of day), the recursive CNN-LSTM is better for only 3 of 28 series in 2025 and 2 of 17 in 2026. For months ahead, climatology is the better estimate.
- Faulty sensor readings were removed before training and scoring. Examples are Gangapur Road NO2 stuck near 3.85 µg/m³ from late August 2025, Gangapur Road NH3 stuck near 12 µg/m³ in 2026, and CO reading 0.00 mg/m³ for months in 2026. Every removed hour is counted and labelled; pollution peaks are never removed.
- 2026 is only a partial test: January-June and August at three stations. July is missing from all files, September-December have not happened in the data, and Pandav Nagari has no 2026 data at all.

Glossary

- Station-pollutant series: the hourly record of one pollutant at one station (4 stations x 7 pollutants = 28 series).
- Persistence baseline: predict that the next hour equals the latest measured value.
- Climatology baseline: predict the historical average for that calendar month and hour of day.
- RMSE / MAE: typical size of the prediction error, in µg/m³ (mg/m³ for CO). Lower is better.
- Skill: 1 - RMSE(model) / RMSE(baseline). Positive means the model beats the baseline; 0.05 means 5% lower error.
- Rolling one-step prediction: each hour is predicted from the real measurements of the previous 24 hours.
- Recursive forecast: the model's own predictions are fed back as inputs, so it can run for a whole year without any future measurement.
- Imputation: filling a missing input value with an estimate. Missing target values are never filled; they are simply not scored.

1. Introduction

Short-term air-quality prediction supports public advisories and operational decisions such as traffic or industrial-emission controls. Continuous ambient monitoring stations report hourly concentrations, but their records contain gaps, instrument faults and changes in data availability, so any forecasting pipeline must handle missing data without using information from after the forecast origin.

Hybrid convolutional-recurrent networks are a common choice for such series: one-dimensional convolutions extract short local patterns, and Long Short-Term Memory (LSTM) layers model longer dependencies [1], [3], [4]. This project applies that design to four Nashik stations, produces an hourly forecast of 2026, and tests the same procedure on 2025, where the full year was measured.

2. Problem Statement

Using only the hourly observations of a station available before 1 January 2026, forecast the concentrations of seven pollutants for every hour of 2026, and measure the accuracy of the same procedure where the answer is known: on 2025, with data up to 31 December 2024, and on the hours of 2026 that have since been measured.

3. Objectives

- Document the dataset's schema, coverage and quality problems per station.
- Build reproducible cleaning and KNN imputation fitted without temporal leakage.
- Train one CNN-LSTM per station on data before each forecast origin.
- Export a continuous hourly 2026 forecast for all four stations, including hours and stations without measurements, without fabricating ground truth.
- Score recursive and one-step predictions separately with MSE, RMSE, MAE and R^2 in original units, against climatology, same-hour-last-year and persistence baselines.
- Derive daily CPCB AQI from the predicted concentrations, clearly separated from measured AQI.

4. Dataset Description

The dataset consists of four CSV files, one per station. Station identity is given only by the file name. All files share 25 columns: a Timestamp plus 24 measurements (criteria pollutants, volatile organic compounds and meteorological variables). The Hirwadi file has an extra Source_File column whose labels do not match the timestamps, so it was dropped and the timestamps were treated as authoritative. Hirwadi rows are stored out of order and were sorted. No file contains duplicate timestamps.

Table 1. Raw files as delivered.

Station	Rows	First timestamp	Last timestamp	Duplicate times	Sorted	Years present	Rows with no pollutant
MIDC Ambad	40152	2017-01-01	2026-08-31 23:00:00	0	True	2017, 2023, 2024, 2025, 2026	14377
Gangapur Road	81024	2017-01-01	2026-08-31	0	True	2017, 2018,	4945

			23:00:00			2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026	
Pandav Nagari	35064	2017-01-01	2025-12-31 23:00:00	0	True	2017, 2023, 2024, 2025	12761
Hirwadi	40152	2017-01-01	2026-08-31 23:00:00	0	False	2017, 2023, 2024, 2025, 2026	13873

Table 2. Coverage of each forecast horizon in the station files.

Station	Horizon	Rows in file (of 8,760)	Hours with a measurement	First measured	Last measured	Months with no rows
MIDC Ambad	2025	8760	8456	2025-01-01 00:00:00	2025-12-31 23:00:00	none
Gangapur Road	2025	8760	8511	2025-01-01 00:00:00	2025-12-31 23:00:00	none
Pandav Nagari	2025	8760	8435	2025-01-01 16:00:00	2025-12-31 23:00:00	none
Hirwadi	2025	8760	8550	2025-01-01 00:00:00	2025-12-31 23:00:00	none
MIDC Ambad	2026	5088	4739	2026-01-01	2026-08-31 23:00:00	Jul, Sep, Oct, Nov, Dec
Gangapur Road	2026	5088	4989	2026-01-01	2026-08-31 23:00:00	Jul, Sep, Oct, Nov, Dec
Pandav Nagari	2026	0	0	-	-	Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec
Hirwadi	2026	5088	4844	2026-01-01	2026-08-31 15:00:00	Jul, Sep, Oct, Nov, Dec

Three files begin with an empty 2017 block and resume in mid-2023; Gangapur Road has data from 2017 onwards. In 2026, every file that has 2026 rows covers 1 January-30 June and 1-31 August (5,088 of 8,760 hours); July and September-December are absent. Pandav Nagari's file ends on 31 December 2025. Its 2026 forecast is produced like the others but cannot be scored.

Seven pollutants are the prediction targets: the pollutants of the Indian National AQI present in the data (PM2.5, PM10, NO2, SO2, CO, O3, NH3). The model uses no weather inputs. The files do contain weather columns, which are measured for about 93-97% of hours at MIDC Ambad, Pandav Nagari and Hirwadi. At Gangapur Road, temperature and humidity cover only about 5% of hours. Weather was left out to keep one consistent input set for all stations.

In January-May 2026 many values at the three stations with 2026 data are reported as whole numbers (25-50% of values, against about 1% before). This looks like a change in reporting precision, not a sensor fault, and these values are kept.

5. Data Preprocessing and KNN Imputation

5.1 Cleaning rules

- Timestamps parsed and sorted; a duplicate timestamp would keep its first row (none occur).
- Rows before a station's first pollutant observation removed (the empty 2017 blocks).
- Negative concentrations set to missing (none occur).

- PM2.5 or PM10 of 999 or more set to missing: the analyser ceiling (21 values). The true value is unknown above the ceiling; one such hour is Diwali night 2024 at Gangapur Road.
- Any value repeated unchanged for 12 or more consecutive hours set to missing as a stuck-sensor reading (13,921 values).
- CO of exactly 0.00 mg/m³ set to missing. Outdoor CO never falls to zero (the clean-air background is about 0.1 mg/m³), and the zeros come in long blocks when the analyser is off (1,645 further values after the stuck rule). Zeros of other pollutants are kept.
- Flatline days set to missing. A failed sensor can jitter around a fixed level instead of repeating one value exactly. For each station, pollutant and calendar day with at least 12 measured hours, the day is removed when the spread of its values (90th minus 10th percentile) is below 5% of its median (40,356 values). Real pollution varies far more within a day, so this rule only removes near-constant days and can never remove a pollution peak.
- No interpolation or resampling. Hours absent from a file are not filled, and no input window spans them.

Table 3. Hourly values removed by each cleaning rule, whole file (all pollutants).

Station	PM ceiling	Stuck 12 h	CO zero	Flatline days
MIDC Ambad	0	3107	84	391
Gangapur Road	21	8482	1241	38626
Pandav Nagari	0	0	9	829
Hirwadi	0	2332	311	510

Table 4. Hourly values removed by the flatline rule, per pollutant (whole file). Gangapur Road dominates: its NO2, NH3, O3 and CO analysers have long near-constant periods (e.g. O3 at 45.0 µg/m³ in December 2021-March 2022, NO2 at 6.28 µg/m³ in March-June 2023, NO2 at 3.85 µg/m³ from late August 2025).

Station	PM2.5	PM10	NO2	SO2	CO	O3	NH3
MIDC Ambad	0	0	0	91	72	85	143
Gangapur Road	14	0	13223	12	5153	5227	14997
Pandav Nagari	48	48	0	168	48	72	445
Hirwadi	0	0	168	111	147	0	84

Detected sensor faults that remain uncertain are reported, not hidden: in mid-2025 CO at Gangapur Road and MIDC Ambad varies only within a few 0.01 mg/m³ steps. These months pass the flatline rule and are kept.

5.2 Missing values

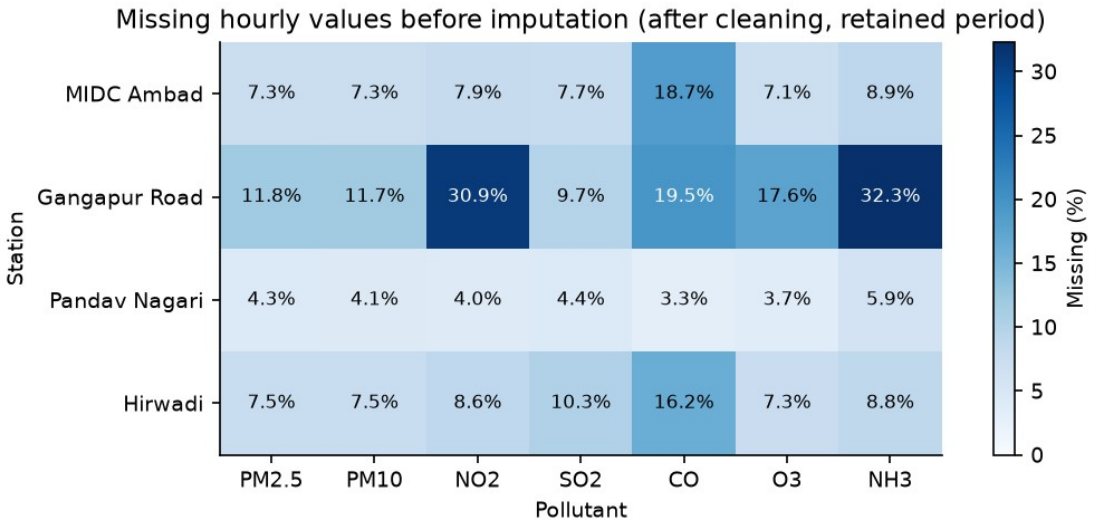


Figure 1. Missing hourly values per station and pollutant before imputation (all data up to the end of the 2026 horizon).

Table 5. Percentage of missing hourly values before imputation.

Station	PM2.5	PM10	NO2	SO2	CO	O3	NH3
MIDC Ambad	7.3	7.3	7.9	7.7	18.7	7.0	8.9
Gangapur Road	11.8	11.7	30.9	9.7	19.5	17.6	32.3
Pandav Nagari	4.3	4.1	4.0	4.4	3.3	3.7	5.9
Hirwadi	7.5	7.5	8.6	10.3	16.2	7.3	8.8

5.3 KNN imputation

The pollutants are first standardised, then imputed. A missing value is filled with the inverse-distance-weighted mean of that variable in the k = 5 most similar hours, using the NaN-Euclidean distance over the seven standardised pollutants of the same hour [2], [5], [6].

- One StandardScaler and one KNNImputer per station and mode, fitted on that station's training rows only, then applied unchanged to validation, test and forecast inputs. Every training row is a candidate donor; no sub-sample is drawn.
- Many missing values occur in hours where all seven pollutants are missing (a whole-station outage). KNN has nothing to compare such an hour with, so it receives the training mean of each pollutant.
- Imputed values are model inputs only. Training targets are never imputed (the loss is masked) and every metric uses measured values only.
- Hours with every pollutant missing get the training mean; windows with more than half of their pollutant cells imputed are discarded.

Table 6. Missing pollutant cells before and after KNN imputation, 2026-forecast mode.

Station	Split	Missing cells before	Missing cells after
MIDC Ambad	train	10148	0

MIDC Ambad	val	1760	0
MIDC Ambad	test	5851	0
Gangapur Road	train	84261	0
Gangapur Road	val	8178	0
Gangapur Road	test	15783	0
Pandav Nagari	train	5178	0
Pandav Nagari	val	1615	0
Pandav Nagari	test	0	0
Hirwadi	train	11734	0
Hirwadi	val	1041	0
Hirwadi	test	5753	0

6. Exploratory Data Analysis

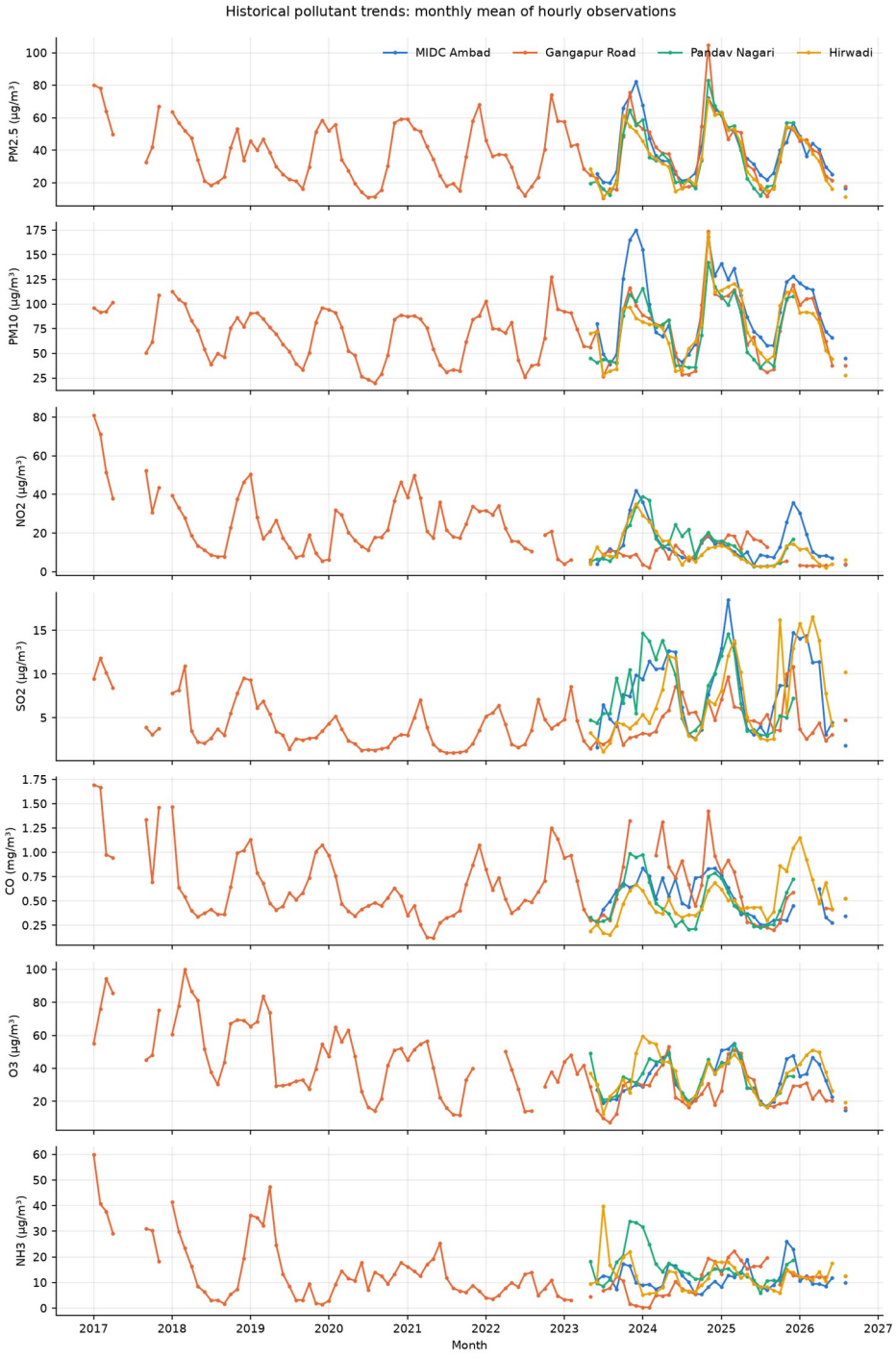


Figure 2. Monthly means of hourly observations per pollutant and station. Gaps are periods with no observations.

Particulate matter follows a seasonal cycle at all stations: low during the monsoon (June-September) and high in winter. This cycle is what a year-ahead forecast can realistically capture. Gangapur Road's gas analysers (NO2, NH3, O3, CO) have long near-constant periods, removed by the flatline rule, and their levels shift between years, so its older history is a weaker guide to 2025-2026.

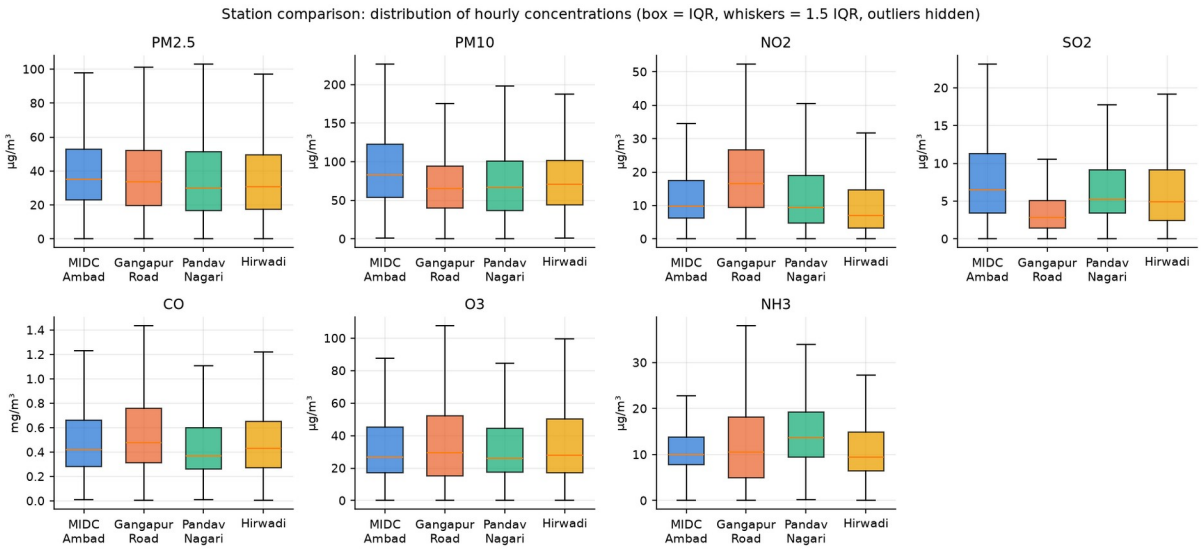


Figure 3. Distribution of hourly concentrations by station (box: interquartile range; whiskers: 1.5 IQR; outliers hidden).

7. CNN-LSTM Architecture

Two causal one-dimensional convolutions (64 filters, kernel 3) extract local temporal motifs from the 24-hour window; max-pooling halves the sequence; an LSTM [1] with 64 units summarises it; dropout [9] regularises; two dense layers output the seven pollutants. Inputs per hour: the seven standardised, KNN-imputed pollutants plus sine and cosine encodings of hour-of-day, day-of-week and month (13 features). Calendar encodings are known for any future hour.

Table 7. CNN-LSTM layers (50,247 trainable parameters per station model).

Layer	Output shape	Configuration	Parameters
Input	(24, 13)	24 hours x 13 features	0
Conv1D	(24, 64)	64 filters, kernel 3, causal padding, ReLU	2,560
Conv1D	(24, 64)	64 filters, kernel 3, causal padding, ReLU	12,352
MaxPooling1D	(12, 64)	pool size 2	0
LSTM	(64)	64 units, final hidden state	33,024
Dropout	(64)	rate 0.2	0
Dense	(32)	ReLU	2,080
Dense (output)	(7)	linear, one unit per pollutant	231

8. Methodology

8.1 Two modes

Table 8. Run configuration. Each mode trains its own models, scalers and imputers and writes to its own output folder.

Mode	Training cutoff	Horizon start	Horizon end	Horizon hours
2025 backtest	2024-12-31 23:00:00	2025-01-01 00:00:00	2025-12-31 23:00:00	8760
2026 forecast	2025-12-31 23:00:00	2026-01-01 00:00:00	2026-12-31 23:00:00	8760

Within each mode, validation is the six months before the forecast origin and training is everything earlier: the 2025 backtest fits the network on data up to 30 June 2024 and validates on July-December 2024; the 2026 run fits on data up to 30 June 2025 and validates on July-December 2025. Scaler, imputer, network weights, early stopping and the choice of target design use only data up to the training cutoff. Horizon rows are used for two things only: as ground truth, and as inputs to the one-step evaluation.

8.2 Two kinds of prediction

- Recursive forecast (the 2026 forecast proper). The model starts from the 24 hours before the origin, predicts the next hour, appends that prediction (with the known calendar features) to its input window, drops the oldest hour and repeats for all 8,760 hours. Only the starting window is real data, so errors accumulate and, beyond the first days, the forecast can only carry the seasonal and diurnal pattern the model has learned. It is compared with three baselines that use the same information: hour-of-day by month climatology, the same hour 52 weeks earlier, and the last value before the origin.
- Rolling one-step evaluation (the task the network was trained for). For each measured horizon hour with a valid window, the frozen model predicts it from the measured previous 24 hours. This needs measurements up to the previous hour, so it exists only where the data exist. It is compared with persistence (the last observed value) on the same rows.

Predicted concentrations are clipped at zero, both before being fed back and on output; a pollutant concentration cannot be negative. Every horizon hour is exported. Each row records whether a measurement exists (observed), a value was rejected by the cleaning rules (removed_by_quality_check), the station file has the hour but no value (missing_value), or the file has no row for the hour (no_row_in_file). Ground truth is never imputed or interpolated, and only observed hours are scored.

8.3 Training and target design

Table 9. Training configuration (same for all stations and modes; no hyperparameter search).

Setting	Value
Loss	Masked MSE on standardised targets (missing targets excluded)
Optimiser	Adam [8], learning rate 0.001
Batch size / maximum epochs	256 / 60
Early stopping	validation loss, patience 6, best weights
Random seed	42 (Python, NumPy, TensorFlow)
Software	Python 3.12.14, TensorFlow 2.19.1, scikit-learn 1.9.1, pandas 3.0.6, NumPy 2.1.3

Two parameterisations of the same network are trained per station: level (predict the standardised concentration) and change (predict the difference from the last hour of the input window, then add it back). Training, one-step and recursive prediction all use that same base value; in the recursive forecast it is the previous prediction. The design for the recursive forecast is chosen by a recursive forecast of the

validation half-year. The design for the one-step evaluation is chosen by one-step validation loss. Neither choice sees the horizon.

Table 10. 2025 backtest: model selection on validation data (MSE, standardised units).

Station	1-step val: persistence	1-step val: level	1-step val: change	Recursive val: level	Recursive val: change	Recursive design	One-step design
MIDC Ambad	0.1928	0.2358	0.1763	2.71	1.615	change	change
Gangapur Road	0.3141	0.2659	0.265	2.915	3.467	level	change
Pandav Nagari	0.2147	0.2356	0.1734	1.325	60.74	level	change
Hirwadi	0.2096	0.2532	0.1694	1.822	3.333	level	change

Table 11. 2026 forecast: model selection on validation data (MSE, standardised units).

Station	1-step val: persistence	1-step val: level	1-step val: change	Recursive val: level	Recursive val: change	Recursive design	One-step design
MIDC Ambad	0.1629	0.1893	0.1369	1.586	2.829	level	change
Gangapur Road	0.143	0.1659	0.1288	2.942	1.484	change	change
Pandav Nagari	0.1219	0.1191	0.09565	1.672	1.742	level	change
Hirwadi	0.3723	0.3762	0.3361	2.683	1.95	change	change

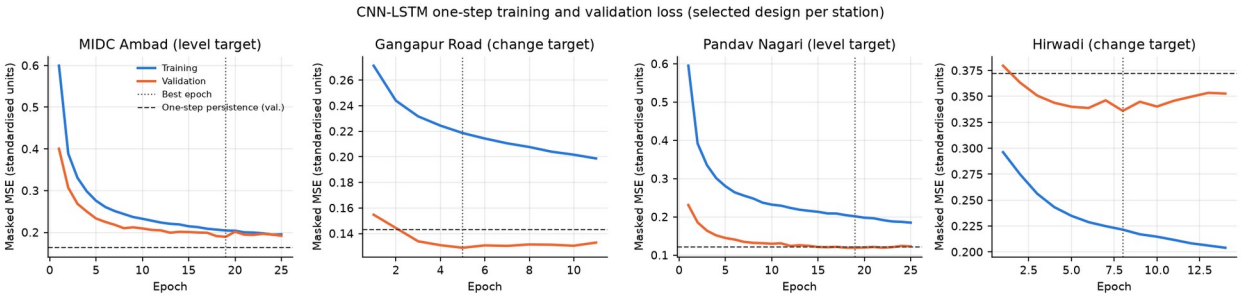


Figure 4. 2026-forecast mode: training and validation loss of the recursively selected design per station.

9. Chronological Splits

Random splitting would score the model on near-copies of its training hours and let it learn from the period it is tested on [10]. Every split here is chronological, and windows are assigned by the timestamp of the predicted hour.

Table 12. Splits per mode and station. Test = the horizon rows present in the file.

Mode	Station	Split	Start	End	Hourly rows	PM2.5 observed
2025 backtest	MIDC Ambad	train	2023-06-14 14:00:00	2024-06-30 23:00:00	9178	8371
2025 backtest	MIDC Ambad	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4054
2025 backtest	MIDC Ambad	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8310
2025 backtest	Gangapur Road	train	2017-01-01 00:00:00	2024-06-30 23:00:00	62760	54175
2025 backtest	Gangapur Road	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4106
2025 backtest	Gangapur Road	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8317
2025 backtest	Pandav Nagari	train	2023-05-22 17:00:00	2024-06-30 23:00:00	9727	9391

2025 backtest	Pandav Nagari	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4251
2025 backtest	Pandav Nagari	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8277
2025 backtest	Hirwadi	train	2023-05-23 16:00:00	2024-06-30 23:00:00	9704	8308
2025 backtest	Hirwadi	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4291
2025 backtest	Hirwadi	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8506
2026 forecast	MIDC Ambad	train	2023-06-14 14:00:00	2025-06-30 23:00:00	17938	16560
2026 forecast	MIDC Ambad	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4175
2026 forecast	MIDC Ambad	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	4708
2026 forecast	Gangapur Road	train	2017-01-01 00:00:00	2025-06-30 23:00:00	71520	62401
2026 forecast	Gangapur Road	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4197
2026 forecast	Gangapur Road	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	4876
2026 forecast	Pandav Nagari	train	2023-05-22 17:00:00	2025-06-30 23:00:00	18487	17736
2026 forecast	Pandav Nagari	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4183
2026 forecast	Pandav Nagari	test	-	-	0	0
2026 forecast	Hirwadi	train	2023-05-23 16:00:00	2025-06-30 23:00:00	18464	16799
2026 forecast	Hirwadi	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4306
2026 forecast	Hirwadi	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	4769

10. Evaluation Metrics

For n measured hours with values y_i, predictions $\hat{y}_i$ and measured mean $\bar{y}$:

$$MSE = (1/n) \Sigma (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE}$$

$$MAE = (1/n) \Sigma |y_i - \hat{y}_i|$$

$$R^2 = 1 - \Sigma (y_i - \hat{y}_i)^2 / \Sigma (y_i - \bar{y})^2$$

- All metrics are in original units, after inverse scaling, on measured hours only. Every method is scored on exactly the same hours.
- n_valid is the number of measured hours scored. Coverage is n_valid as a share of the 8,760 horizon hours.
- R^2 is reported as undefined with fewer than two hours or zero variance, and flagged unreliable with fewer than 168 hours (one week).
- Recursive skill = 1 − RMSE_CNN-LSTM / RMSE_climatology; one-step skill = 1 − RMSE_CNN-LSTM / RMSE_persistence. Positive means the CNN-LSTM is better.
- Hours removed by the cleaning rules are never scored, for any method.

- As a second safeguard, a series whose remaining measured horizon values take fewer than 100 distinct values is quality-flagged. It is reported but left out of the pooled per-pollutant summary and the win counts. Pollutants are never pooled together.
- R^2 is never read on its own: hourly pollution is persistent, so even the persistence baseline reaches a high R^2. The skill against a baseline is the number that shows whether the model adds anything.

11. Results: 2025 Backtest

Models trained on data up to 31 December 2024 forecast every hour of 2025, which is measured at all four stations. This is the fair test of how the procedure behaves on a year it has never seen.

11.1 Recursive year-ahead forecast

*Table 13. 2025: recursive forecast per station and pollutant, measured hours only (µg/m³; CO in mg/m³). * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	Cov. %	RMSE	MAE	R^2	RMSE clim.	RMSE last yr	RMSE persist.	Skill vs clim.	R^2 flag
MIDC Ambad	PM2.5	8310	94.86	125.8	105.4	-30.56	20.92	26.46	24.06	-5.013	ok
MIDC Ambad	PM10	8327	95.06	61.53	47.47	-0.3626	51.64	63.78	53.07	-0.1916	ok
MIDC Ambad	NO2	8267	94.37	25.63	20.42	-2.976	11.83	15.85	12.97	-1.166	ok
MIDC Ambad	SO2	8275	94.46	17.68	14.66	-3.233	8.281	10.03	8.887	-1.135	ok
MIDC Ambad	CO	8286	94.59	0.8694	0.6989	-8.762	0.3626	0.575	0.4218	-1.398	ok
MIDC Ambad	O3	8415	96.06	25.85	19.32	-0.1983	19.33	23.47	24.11	-0.3372	ok
MIDC Ambad	NH3	8079	92.23	12.84	9.125	-0.8606	9.615	10.79	12.09	-0.3353	ok
Gangapur Road	PM2.5	8317	94.94	34.79	28.09	-0.6057	22.13	36.05	49.31	-0.5721	ok
Gangapur Road	PM10	8276	94.47	58.51	44.08	-0.1657	43.94	60.57	76.48	-0.3316	ok
Gangapur Road	NO2	4967	56.7	24.92	19.8	-9.095	15.07	13.5	14.04	-0.6543	ok
Gangapur Road	SO2	8438	96.32	9.191	6.088	-0.7786	6.758	8.121	8.268	-0.3601	ok
Gangapur Road	CO	8041	91.79	0.4873	0.3703	-0.6611	0.4279	0.7289	1.358	-0.1387	ok
Gangapur Road	O3	7911	90.31	82.71	67.88	-16.4	24.36	23.11	27.52	-2.395	ok
Gangapur Road	NH3	4984	56.89	13.09	9.985	-1.992	9.437	14.56	9.44	-0.3869	ok
Pandav Nagari	PM2.5	8277	94.49	30.81	20.26	-0.3628	21.18	30.79	26.42	-0.4548	ok
Pandav Nagari	PM10	8309	94.85	49.91	36.35	-0.2096	36.13	53.96	46.26	-0.3816	ok
Pandav Nagari	NO2	8347	95.29	11.11	7.995	-0.6995	14.2	19.26	8.618	0.218	ok
Pandav Nagari	SO2	8301	94.76	9.574	4.119	0.008285	9.121	11.78	9.688	-0.04967	ok
Pandav Nagari	CO	8407	95.97	0.3095	0.229	0.06024	0.2568	0.3219	0.3194	-0.205	ok
Pandav Nagari	O3	8350	95.32	24.23	18.24	-0.3169	16.31	21.47	42.69	-0.4854	ok
Pandav Nagari	NH3	8205	93.66	14.17	10.7	-2.806	9.382	11.85	7.515	-0.5104	ok
Hirwadi	PM2.5	8506	97.1	17.96	12.56	0.4253	19.27	25.78	27.81	0.06797	ok
Hirwadi	PM10	8494	96.96	43.65	29.09	0.213	44.96	61.58	49.38	0.02908	ok
Hirwadi	NO2	8503	97.07	18.1	14.33	-5.203	11.41	13.77	7.868	-0.5868	ok
Hirwadi	SO2	8147	93	12.77	8.097	-0.4551	10.91	11.66	12.53	-0.1707	ok
Hirwadi	CO	8527	97.34	0.3402	0.2461	-0.01539	0.2967	0.3598	0.3377	-0.1466	ok
Hirwadi	O3	8528	97.35	22.94	17.64	-0.01753	15.73	21.27	22.77	-0.4585	ok
Hirwadi	NH3	8472	96.71	24.27	16.06	-6.666	10.9	10.48	9.079	-1.227	ok

Table 14. 2025: recursive forecast, unflagged stations pooled per pollutant.

Pollutant	n	MSE	RMSE	MAE	R^2	RMSE clim.	Skill vs clim.	Stations
PM2.5	33410	4,554	67.48	41.41	-6.231	20.89	-2.23	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
PM10	33406	2,896	53.82	39.19	-0.09849	44.52	-0.2089	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NO2	30084	409.9	20.25	15.15	-3.064	12.99	-0.5588	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
SO2	33161	162.5	12.75	8.228	-0.9765	8.876	-0.4361	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
CO	33261	0.2996	0.5474	0.3846	-1.658	0.341	-0.6052	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
O3	33204	2,082	45.63	30.19	-3.293	19.14	-1.383	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NH3	29740	296.7	17.22	11.68	-3.02	9.907	-0.7386	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi

Pooled RMSE skill against climatology ranges from -223.0% (PM2.5) to -20.9% (PM10): the CNN-LSTM beats climatology for 0 of 7 pollutants. A year-ahead recursive forecast cannot know the weather of a given day, so after the first hours its value lies in reproducing the seasonal and daily cycle, and climatology is the baseline it has to match. It does not: the CNN-LSTM beats climatology for only 3 of 28 station-pollutant series. Feeding predictions back lets the network settle on its own level, which can be far from the measurements for months. The worst series is MIDC Ambad PM2.5 (RMSE 125.8 against 20.9 for climatology).

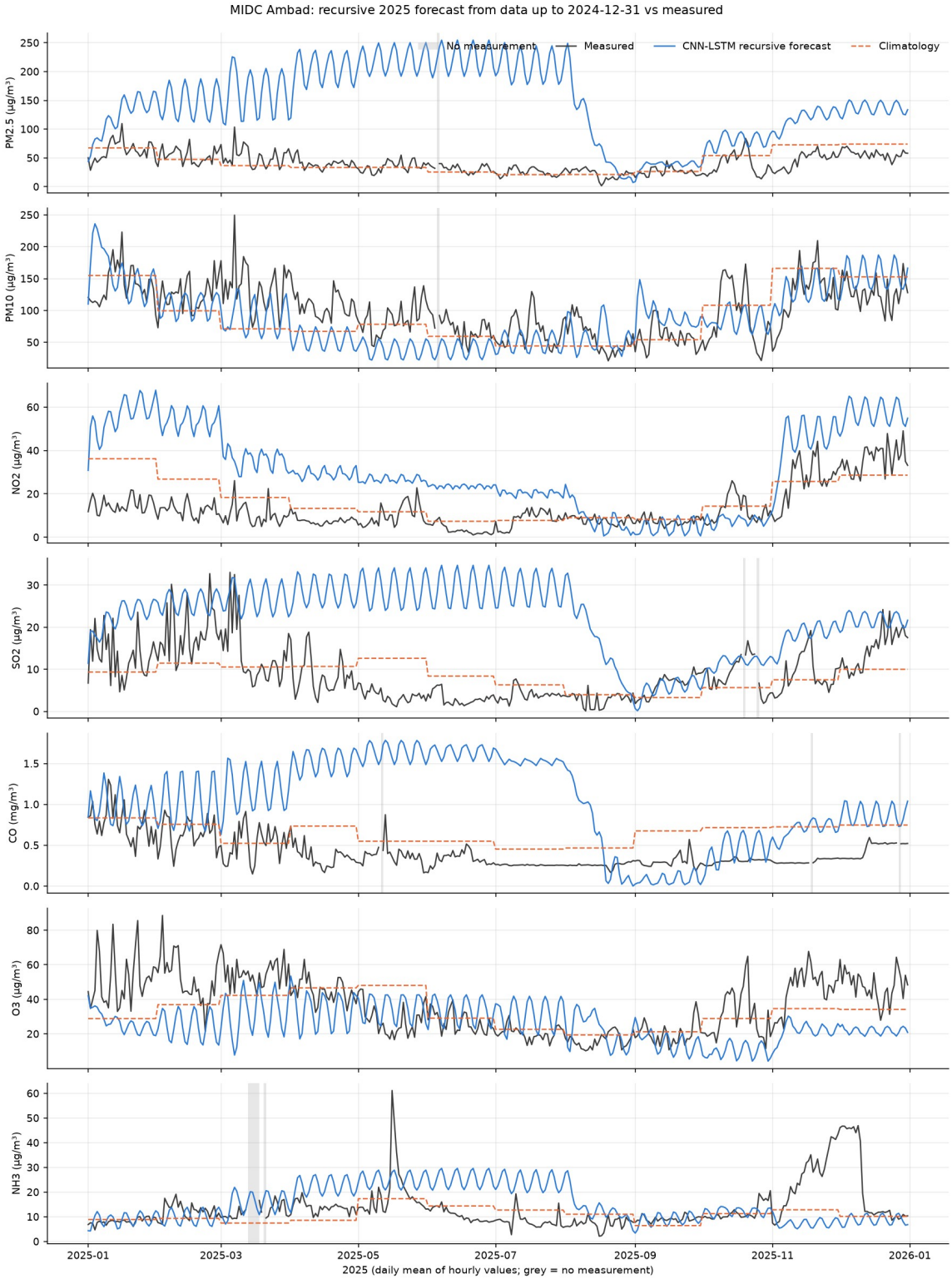


Figure 5. MIDC Ambad: 2025 recursive forecast (daily means) against measurements.

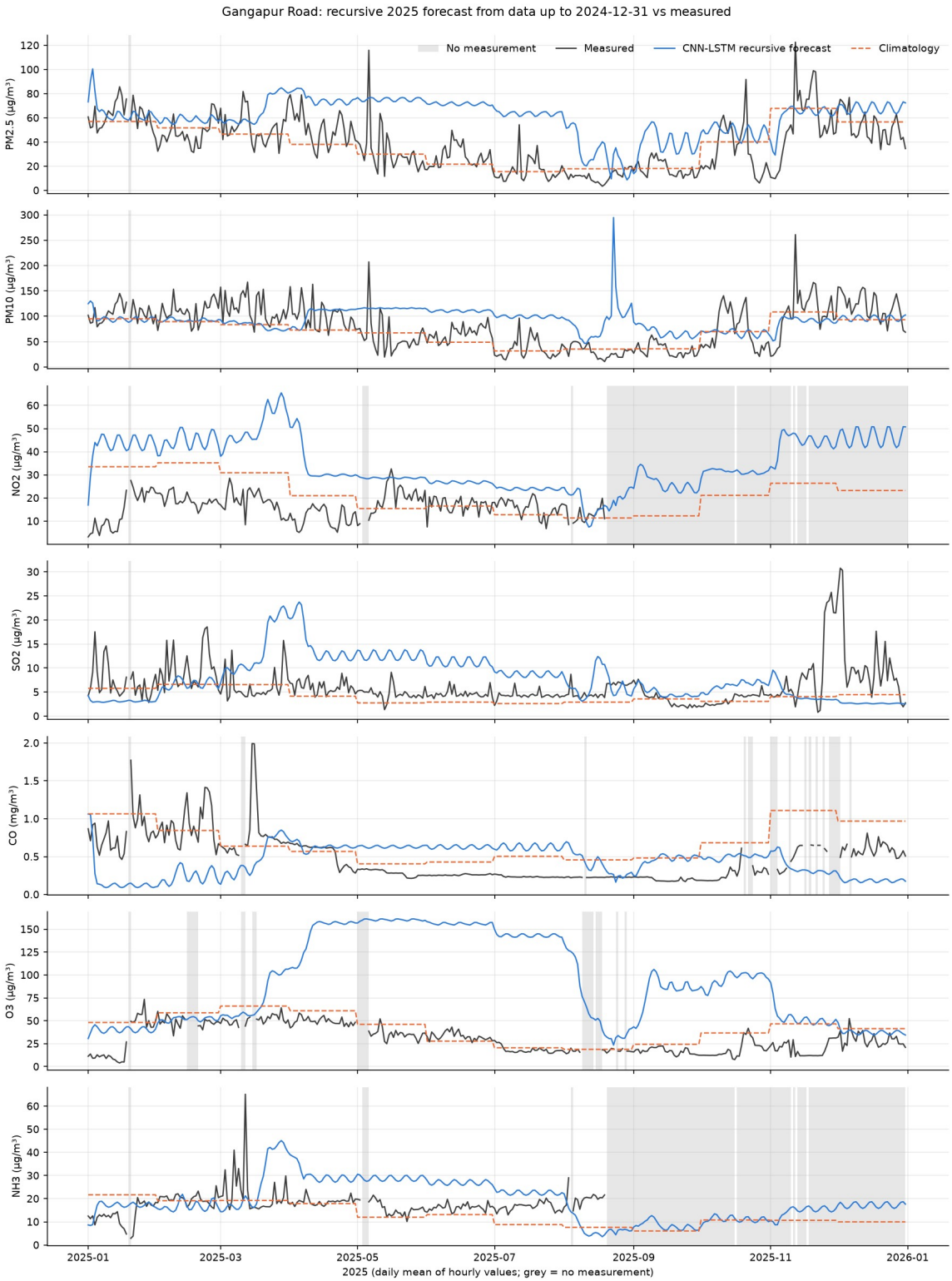


Figure 6. Gangapur Road: 2025 recursive forecast (daily means) against measurements.

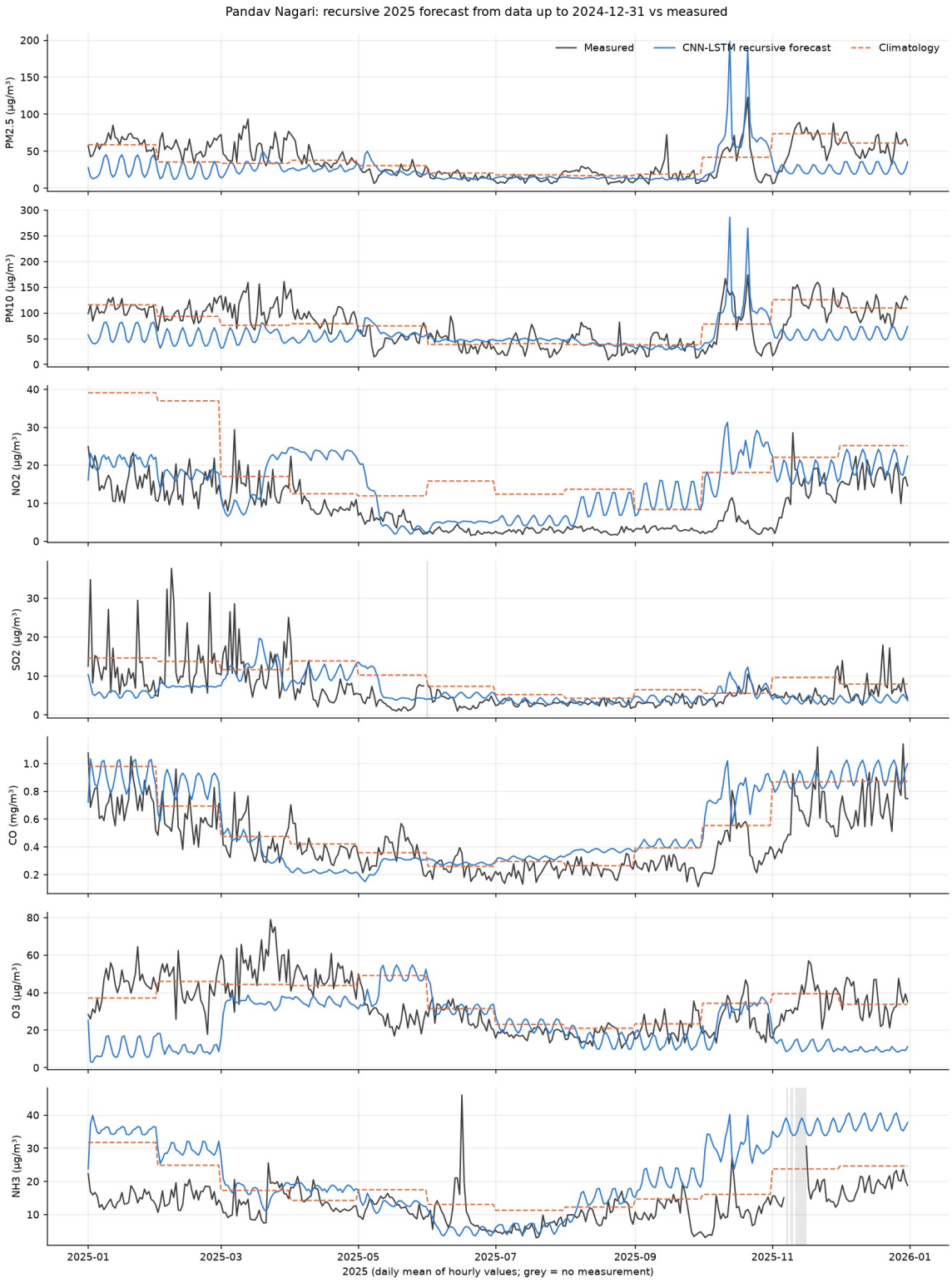


Figure 7. Pandav Nagari: 2025 recursive forecast (daily means) against measurements.

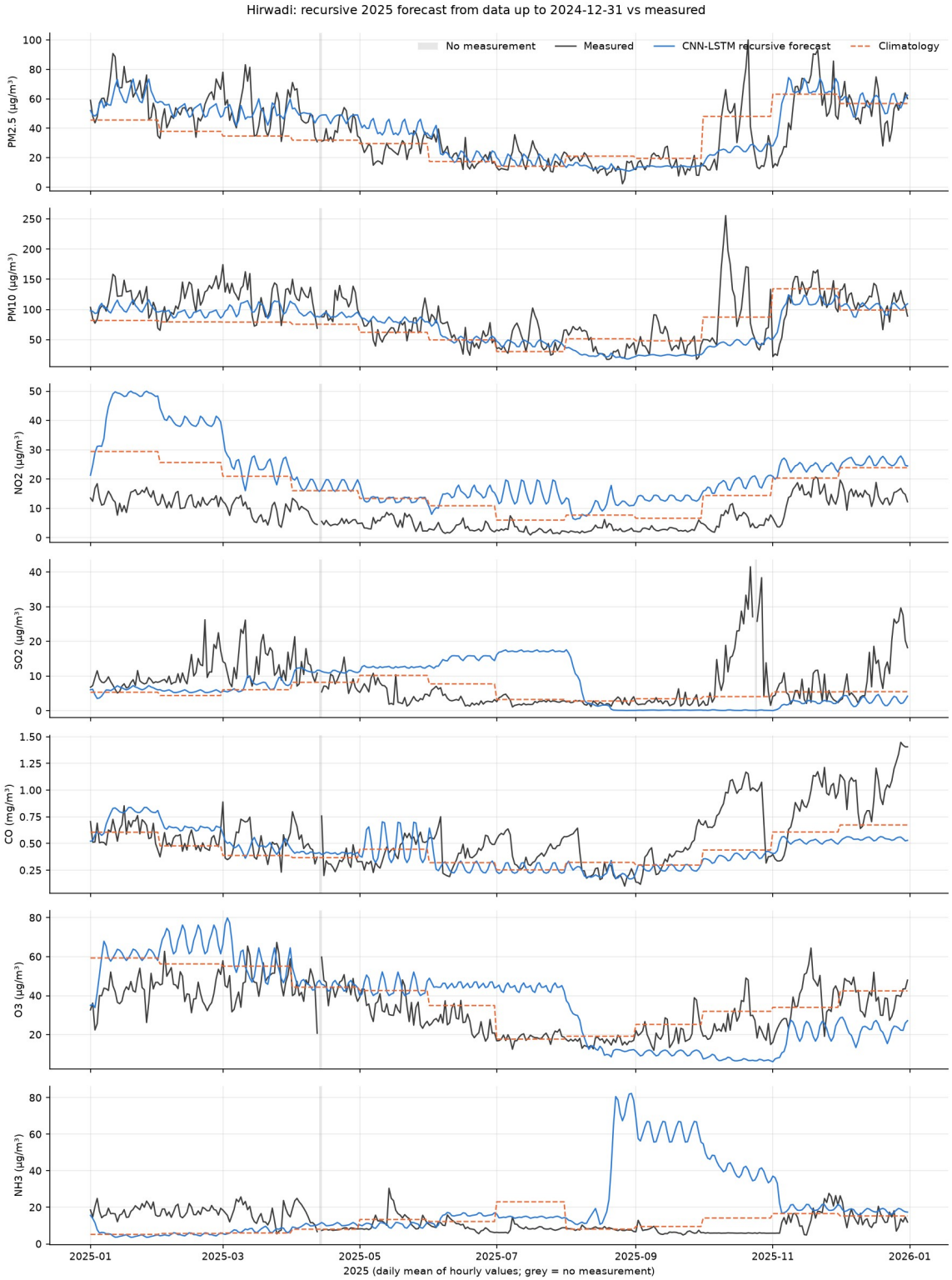


Figure 8. Hirwadi: 2025 recursive forecast (daily means) against measurements.

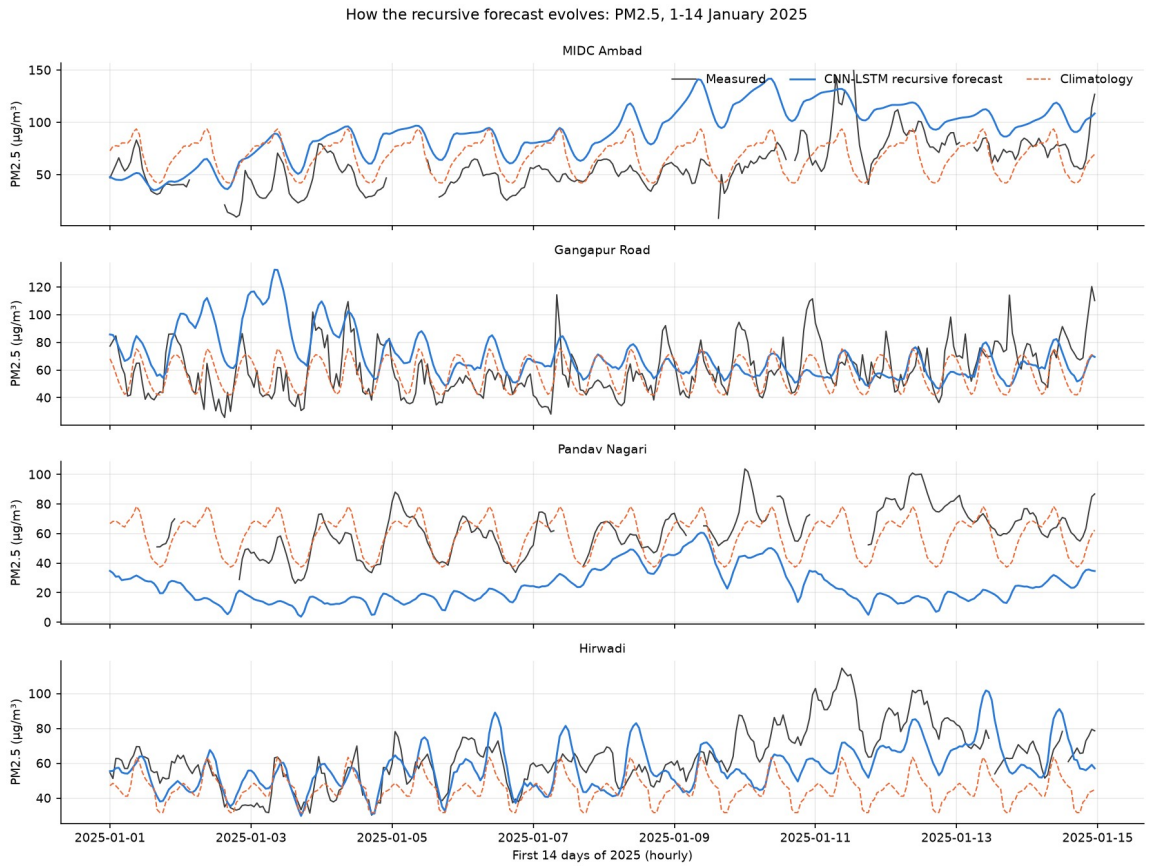


Figure 9. PM2.5 in the first 14 days of 2025: the recursive forecast drifts from the measurements within days.

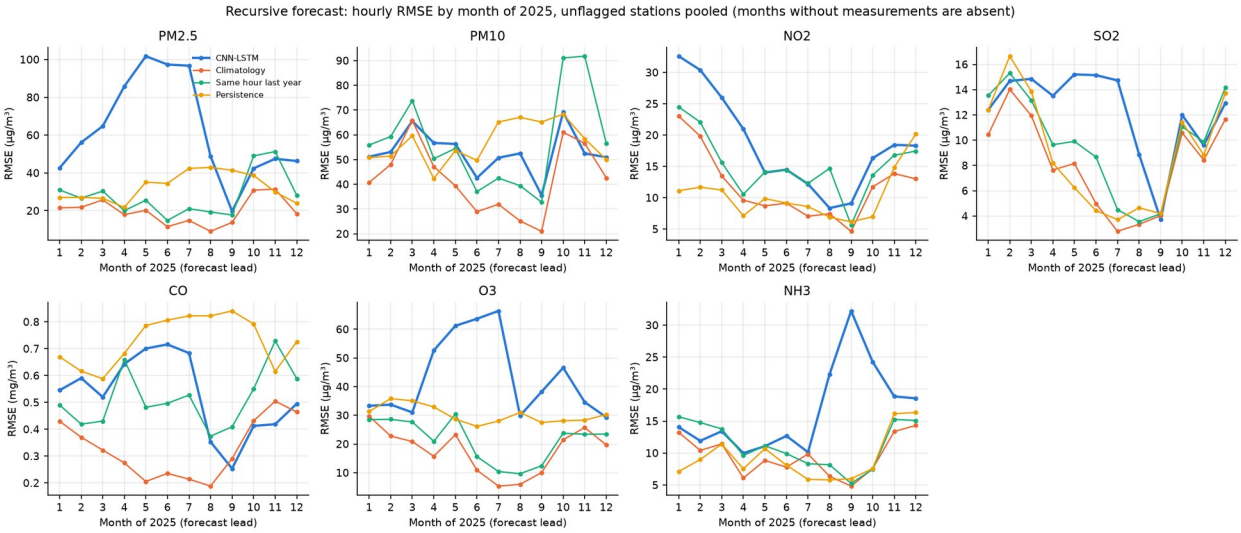


Figure 10. Recursive forecast RMSE by month of 2025 (forecast lead), unflagged stations pooled.

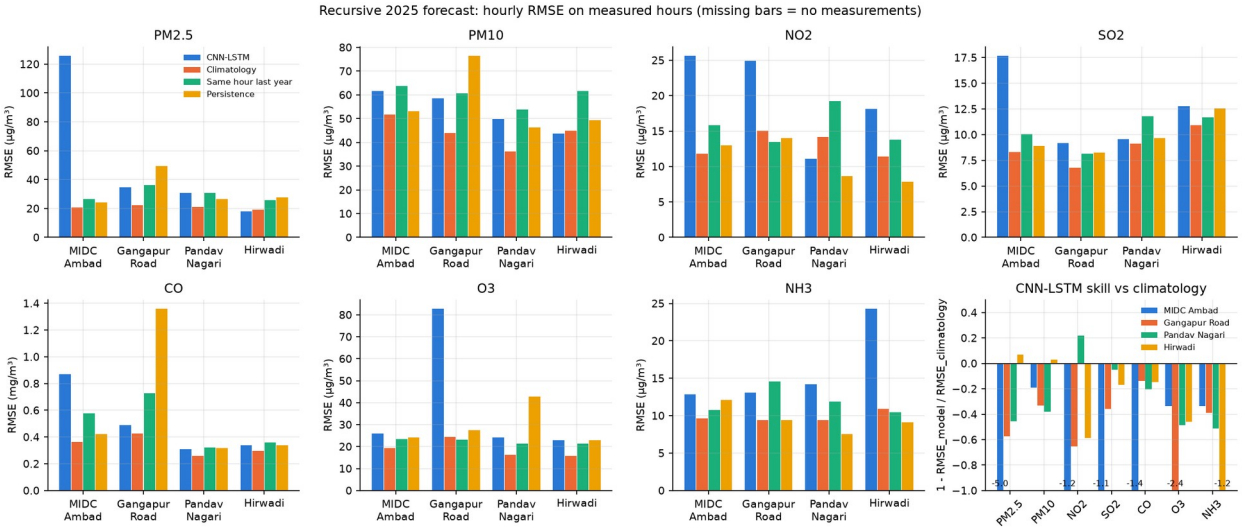


Figure 11. 2025 recursive RMSE per station and baseline, and RMSE skill against climatology.

11.2 Rolling one-step evaluation

*Table 15. 2025: rolling one-step predictions per station and pollutant. * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	MSE	RMSE	MAE	R^2	RMSE persist.	Skill vs persist.	R^2 flag
MIDC Ambad	PM2.5	8231	69.1	8.313	5.031	0.8628	9.28	0.1042	ok
MIDC Ambad	PM10	8248	541.4	23.27	11.24	0.8055	25.57	0.09009	ok
MIDC Ambad	NO2	8196	37.71	6.141	3.587	0.7694	6.565	0.06468	ok
MIDC Ambad	SO2	8197	26.65	5.163	1.986	0.6392	5.383	0.04093	ok
MIDC Ambad	CO	8212	0.03399	0.1844	0.09637	0.5616	0.1827	-0.009157	ok
MIDC Ambad	O3	8338	77.16	8.784	5.73	0.861	9.702	0.09464	ok
MIDC Ambad	NH3	8009	7.653	2.766	1.323	0.9139	2.641	-0.04753	ok
Gangapur Road	PM2.5	8179	227.3	15.08	5.668	0.6936	16.05	0.06057	ok
Gangapur Road	PM10	8145	683.7	26.15	12.41	0.7636	28.13	0.07065	ok
Gangapur Road	NO2	4932	46.72	6.835	4.852	0.2345	6.134	-0.1143	ok
Gangapur Road	SO2	8272	24.89	4.989	2.212	0.4815	5.746	0.1318	ok
Gangapur Road	CO	7885	0.03546	0.1883	0.08974	0.7503	0.1958	0.03808	ok
Gangapur Road	O3	7747	66.98	8.184	4.529	0.8305	7.12	-0.1495	ok
Gangapur Road	NH3	4953	37.06	6.088	3.095	0.3327	6.136	0.007934	ok
Pandav Nagari	PM2.5	8134	85.51	9.247	3.989	0.8768	10.26	0.09884	ok
Pandav Nagari	PM10	8160	239.7	15.48	7.733	0.8825	16.38	0.05471	ok
Pandav Nagari	NO2	8198	16.6	4.075	2.25	0.7673	4.291	0.05029	ok
Pandav Nagari	SO2	8152	32.81	5.728	2.096	0.6344	6.496	0.1182	ok
Pandav Nagari	CO	8258	0.01929	0.1389	0.07531	0.8086	0.1585	0.1238	ok
Pandav Nagari	O3	8201	52.06	7.215	4.664	0.8809	8.28	0.1286	ok
Pandav Nagari	NH3	8063	17.11	4.136	1.874	0.674	4.955	0.1652	ok
Hirwadi	PM2.5	8469	61.72	7.856	4.437	0.8901	7.686	-0.02218	ok
Hirwadi	PM10	8457	560.3	23.67	9.619	0.7685	24.1	0.01782	ok
Hirwadi	NO2	8466	24.08	4.907	3.173	0.5453	3.9	-0.2584	ok
Hirwadi	SO2	8111	66.39	8.148	3.104	0.4096	8.627	0.0555	ok
Hirwadi	CO	8490	0.02721	0.165	0.07528	0.7615	0.1757	0.06144	ok
Hirwadi	O3	8491	94.03	9.697	5.921	0.8181	10.96	0.1149	ok
Hirwadi	NH3	8435	40.77	6.385	2.564	0.471	6.542	0.0239	ok

Table 16. 2025: one-step predictions, unflagged stations pooled per pollutant.

Pollutant	n	RMSE	R^2	RMSE persist.	Skill vs persist.	Stations
PM2.5	33013	10.51	0.8238	11.24	0.06516	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
PM10	33010	22.51	0.807	23.95	0.06024	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NO2	29792	5.433	0.705	5.242	-0.03659	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
SO2	32732	6.131	0.5411	6.675	0.08148	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
CO	32845	0.17	0.7427	0.1784	0.04747	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
O3	32777	8.535	0.8492	9.175	0.06975	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NH3	29460	4.967	0.6651	5.215	0.04766	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi

With the measured previous 24 hours as input, pooled RMSE skill against persistence ranges from -3.7% (NO2) to 8.1% (SO2). Persistence is a strong baseline one hour ahead because consecutive hourly averages are highly correlated.

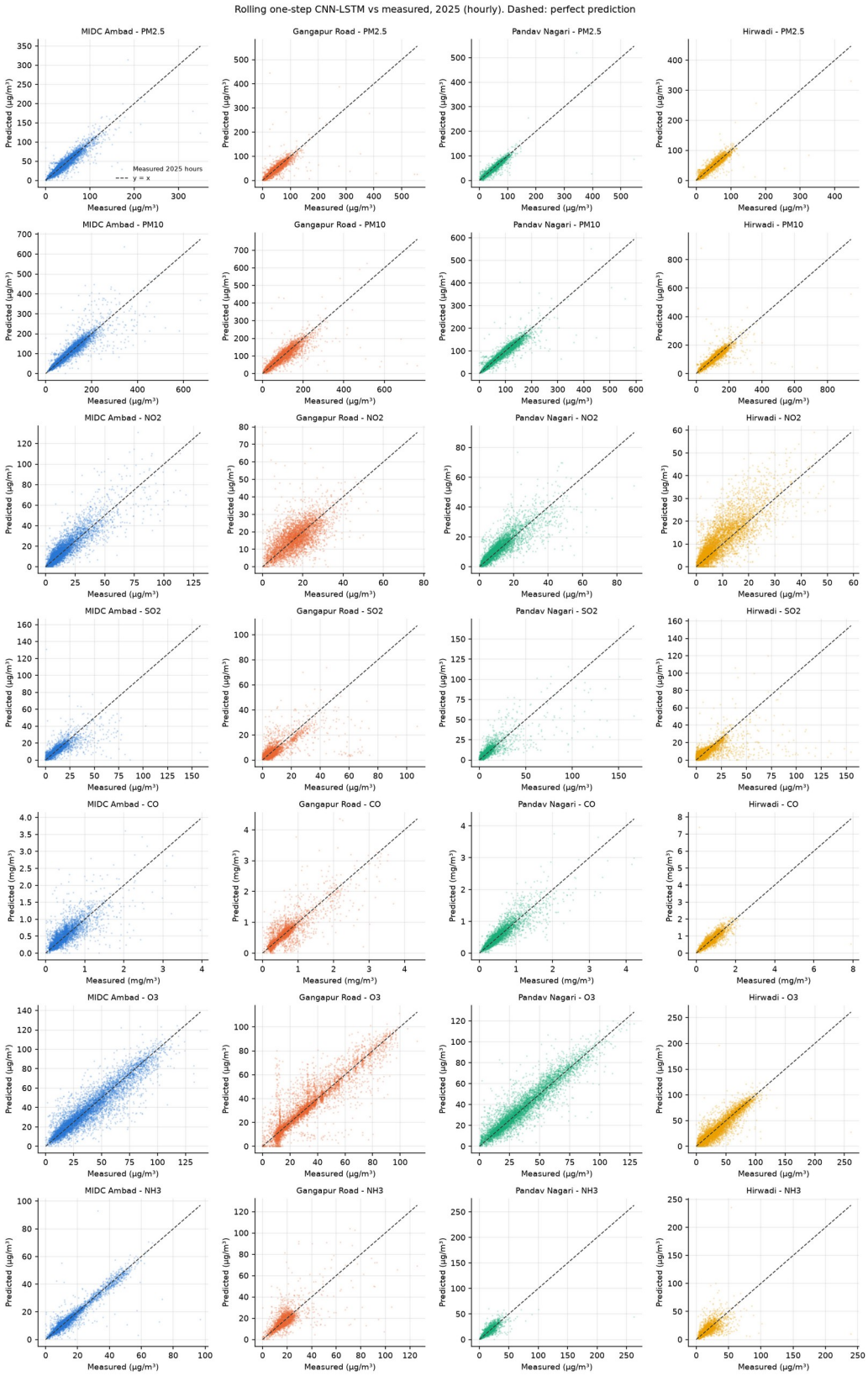


Figure 12. 2025 one-step predictions against measurements. Dashed: perfect prediction.

12. Results: 2026 (Partial-Year Evaluation)

Models trained on data up to 31 December 2025 forecast every hour of 2026 at all four stations (8,760 hours each, 1 January 00:00 to 31 December 23:00). The forecast file contains every hour. Only hours labelled observed are scored: January-June and August at MIDC Ambad, Gangapur Road and Hirwadi. July and September-December 2026, and all of 2026 at Pandav Nagari, are forecasts without ground truth. Because those months are missing, the 2026 scores do not cover a full seasonal cycle (the monsoon months July-September are mostly absent).

12.1 Plausibility of the unmeasured months

Metrics cannot judge the months without measurements, so Table 17 compares the monthly mean PM2.5 forecast with the station's climatology for every month of 2026. Large ratios in either direction mark drift of the recursive forecast rather than an expected change in air quality.

Table 17. 2026 monthly mean PM2.5 (µg/m³): recursive CNN-LSTM forecast against climatology.

Station	Series	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
MIDC Ambad	Forecast	52	51	48	31	41	36	29	30	38	49	64	66
MIDC Ambad	Climatology	65	50	44	37	34	28	22	21	26	50	63	68
Gangapur Road	Forecast	68	60	60	56	42	11	3	4	4	11	21	13
Gangapur Road	Climatology	58	51	47	40	30	23	16	17	18	39	66	56
Pandav Nagari	Forecast	82	94	84	29	19	16	15	15	21	28	119	135
Pandav Nagari	Climatology	60	44	45	39	27	19	16	17	19	40	68	60
Hirwadi	Forecast	90	100	102	98	93	90	91	95	91	64	14	18
Hirwadi	Climatology	56	45	44	38	28	19	16	18	18	44	60	56

Months where the forecast is more than twice, or less than half of, climatology: Gangapur Road (Jun, Jul, Aug, Sep, Oct, Nov, Dec; change design); Hirwadi (Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Nov, Dec; change design); Pandav Nagari (Feb, Dec; level design). The most extreme is Gangapur Road in July: 3 µg/m³ against a climatology of 16. Both target designs drift: each prediction becomes an input for the next, so small biases accumulate over thousands of steps and carry the forecast away from realistic levels. These values, and the AQI derived from them, should not be used as an estimate of 2026 air quality; the climatology column of the same file is the better year-ahead estimate.

12.2 Recursive forecast on measured hours

*Table 18. 2026: recursive forecast per station and pollutant, measured hours only (µg/m³; CO in mg/m³). * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	Cov. %	RMSE	MAE	R^2	RMSE clim.	RMSE last yr	RMSE persist.	Skill vs clim.	R^2 flag
MIDC Ambad	PM2.5	4708	53.74	16.85	13.32	-0.05773	15.78	23.89	21.07	-0.06809	ok
MIDC Ambad	PM10	4710	53.77	37.63	27.26	0.1383	31.89	50.19	42.71	-0.1799	ok
MIDC	NO2	4629	52.84	21.78	17.55	-1.941	8.678	12	16.36	-1.51	ok

Ambad											
MIDC Ambad	SO2	4708	53.74	7.902	6.101	-0.4324	5.131	9.114	10.74	-0.54	ok
MIDC Ambad	CO *	1638	18.7	0.7627	0.6727	-49.96	0.1734	0.1731	0.2145	-3.4	ok
MIDC Ambad	O3	4718	53.86	24.88	19.06	-0.6393	13.97	21.99	20.21	-0.7809	ok
MIDC Ambad	NH3	4654	53.13	13.53	9.949	-23.49	4.64	7.411	3.002	-1.916	ok
Gangapur Road	PM2.5	4876	55.66	25.22	20.58	-0.3836	19.44	29.13	22.04	-0.297	ok
Gangapur Road	PM10	4856	55.43	38.48	29.89	0.1962	32.53	51.12	43.41	-0.183	ok
Gangapur Road	NO2 *	490	5.594	29.35	26.23	-317.6	25.92	15.43	1.787	-0.1324	ok
Gangapur Road	SO2	4317	49.28	5.452	3.811	-1.907	4.519	8.16	3.226	-0.2065	ok
Gangapur Road	CO *	87	0.9932	0.6562	0.5964	-185.8	0.1186	0.2722	0.04945	-4.532	unreliable (n < 168)
Gangapur Road	O3	4971	56.75	39.62	31.22	-8.208	36.55	30.51	18.08	-0.08403	ok
Gangapur Road	NH3 *	236	2.694	11.19	11.1	-2,485	6.585	6.092	0.4801	-0.7	ok
Pandav Nagari	PM2.5	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	PM10	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	NO2	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	SO2	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	CO	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	O3	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	NH3	0	0	-	-	-	-	-	-	-	undefined
Hirwadi	PM2.5	4769	54.44	67.05	64.66	-14.15	12.16	20.42	23.74	-4.516	ok
Hirwadi	PM10	4767	54.42	159.1	156	-18.29	28.6	47.93	38.32	-4.561	ok
Hirwadi	NO2	4352	49.68	7.35	5.828	-0.3297	8.726	6.239	6.822	0.1577	ok
Hirwadi	SO2	4549	51.93	24.81	22.34	-3.087	12.56	14.42	13.94	-0.9747	ok
Hirwadi	CO	2099	23.96	0.2942	0.2353	0.1651	0.3293	0.3813	0.7859	0.1065	ok
Hirwadi	O3	4779	54.55	30.51	25.85	-0.5059	16.94	20.79	25.13	-0.8007	ok
Hirwadi	NH3	4548	51.92	27.83	24.58	-8.533	9.265	12.2	12.17	-2.004	ok

Table 19. 2026: recursive forecast, unflagged stations pooled per pollutant.

Pollutant	n	MSE	RMSE	MAE	R^2	RMSE clim.	Skill vs clim.	Stations
PM2.5	14353	1,803	42.46	32.85	-4.214	16.1	-1.638	MIDC Ambad, Gangapur Road, Hirwadi
PM10	14333	9,383	96.87	70.98	-4.595	31.06	-2.118	MIDC Ambad, Gangapur Road, Hirwadi
NO2	8981	270.7	16.45	11.87	-1.424	8.701	-0.8907	MIDC Ambad, Hirwadi
SO2	13574	237.4	15.41	10.81	-1.944	8.278	-0.8613	MIDC Ambad, Gangapur Road, Hirwadi
CO	2099	0.08657	0.2942	0.2353	0.1651	0.3293	0.1065	Hirwadi
O3	14468	1,049	32.38	25.48	-1.44	24.85	-0.3032	MIDC Ambad, Gangapur Road, Hirwadi
NH3	9202	475.5	21.81	17.18	-9.46	7.301	-1.986	MIDC Ambad, Hirwadi

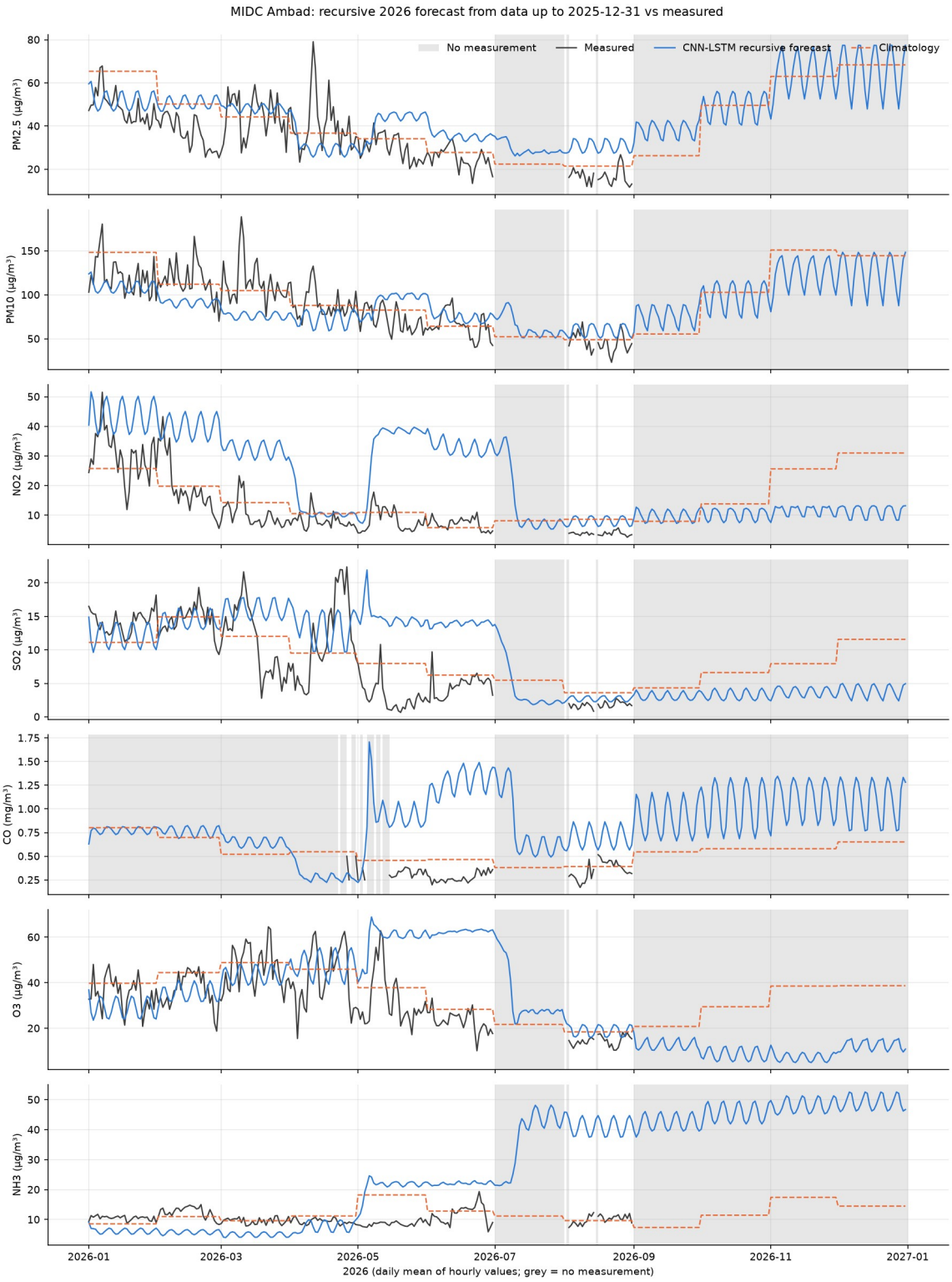


Figure 13. MIDC Ambad: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

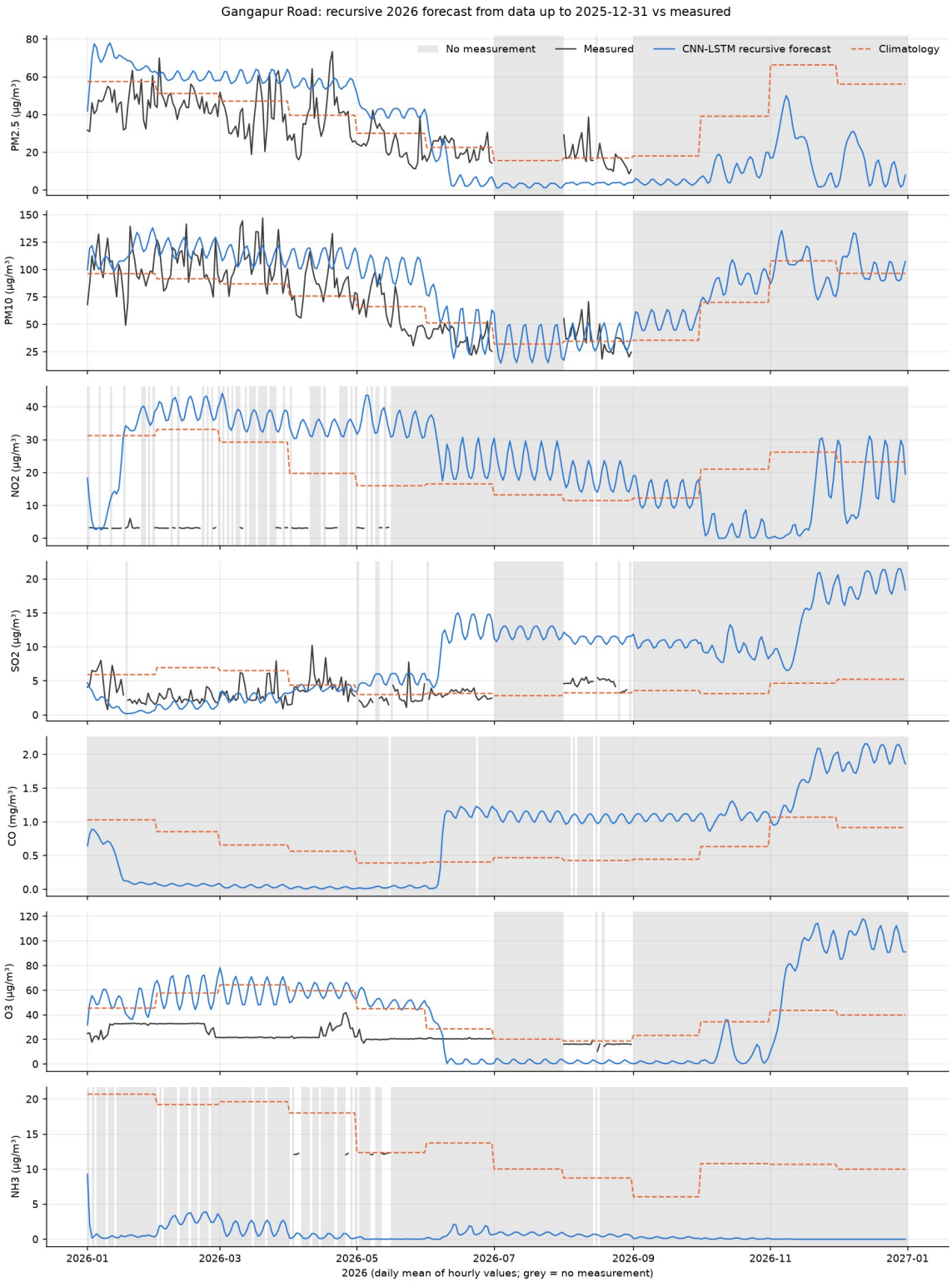


Figure 14. Gangapur Road: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

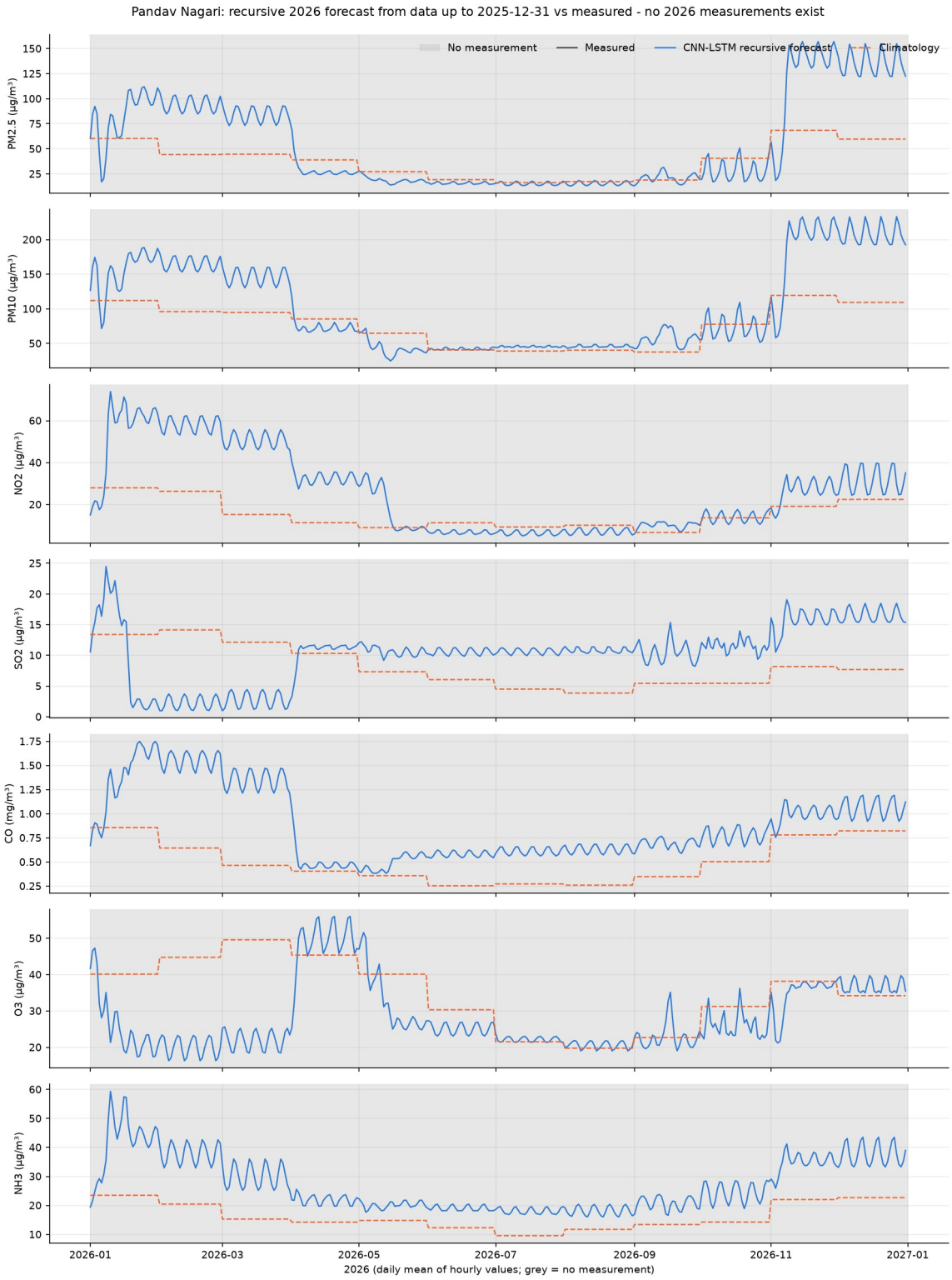


Figure 15. Pandav Nagari: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

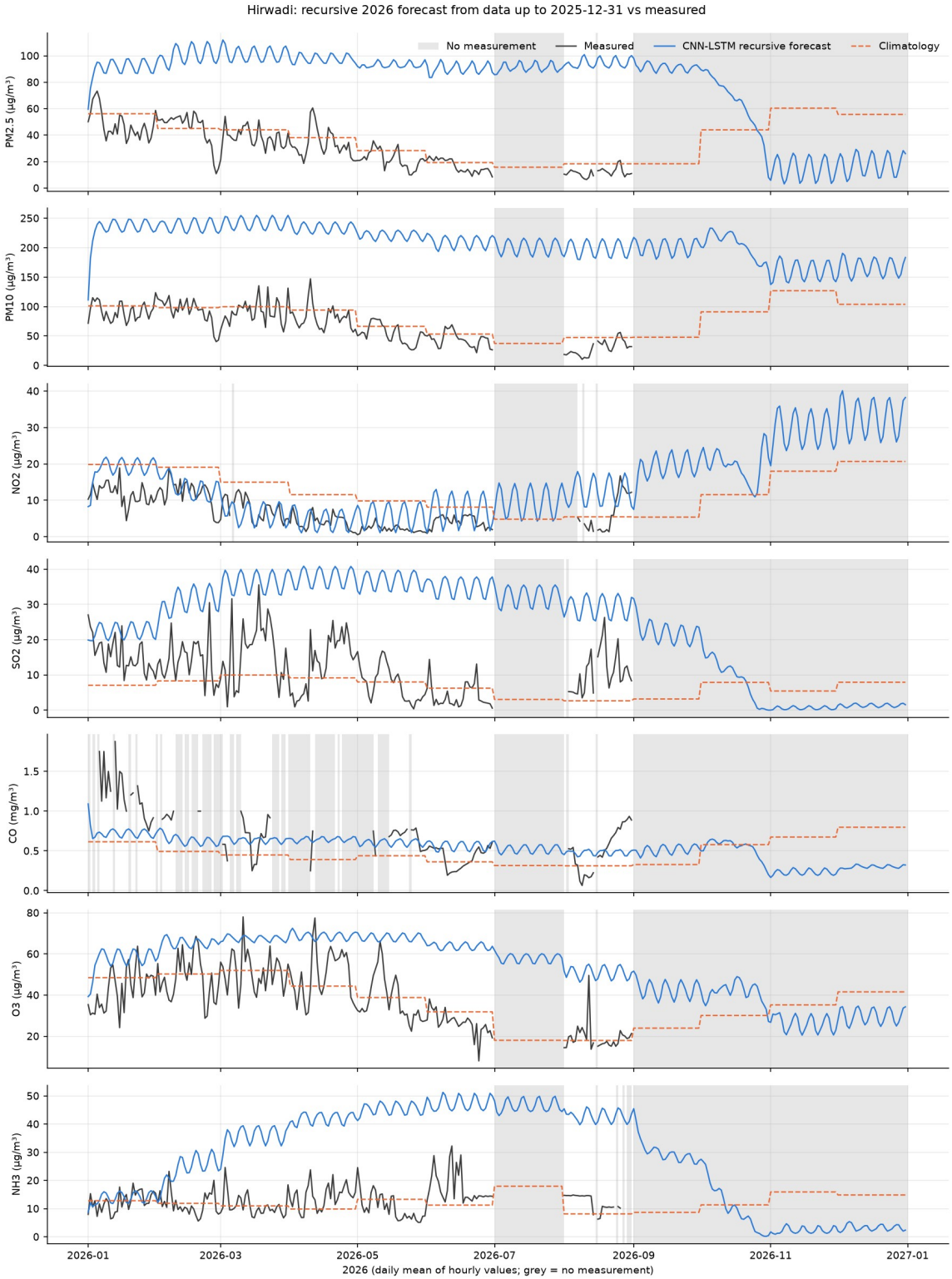


Figure 16. Hirwadi: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

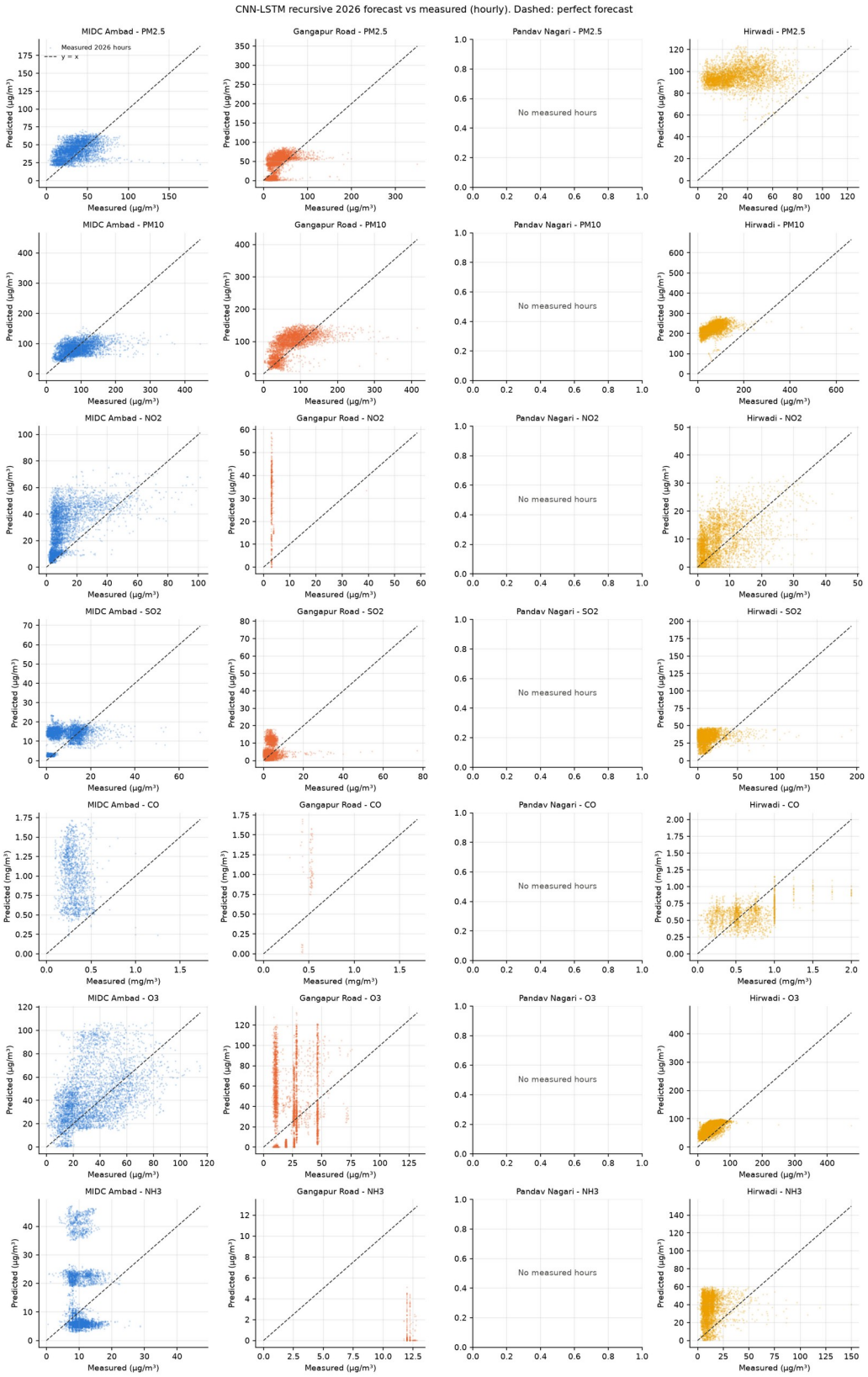


Figure 17. 2026 recursive forecast against measured hourly values.

12.3 Rolling one-step evaluation on measured hours

*Table 20. 2026: rolling one-step predictions per station and pollutant. * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	MSE	RMSE	MAE	R^2	RMSE persist.	Skill vs persist.	R^2 flag
MIDC Ambad	PM2.5	4542	34.33	5.859	3.908	0.8708	6.106	0.04052	ok
MIDC Ambad	PM10	4544	238.1	15.43	8.791	0.8547	17.98	0.1417	ok
MIDC Ambad	NO2	4467	24.48	4.948	2.788	0.8488	5.958	0.1696	ok
MIDC Ambad	SO2	4542	5.328	2.308	1.257	0.8731	2.496	0.07536	ok
MIDC Ambad	CO *	1543	0.00804	0.08967	0.05258	0.2773	0.07957	-0.1268	ok
MIDC Ambad	O3	4552	43.13	6.567	4.565	0.8861	7.68	0.1448	ok
MIDC Ambad	NH3	4489	1.933	1.39	0.904	0.7443	1.378	-0.009133	ok
Gangapur Road	PM2.5	4267	159.8	12.64	5.655	0.6674	12.63	-0.0007244	ok
Gangapur Road	PM10	4267	367.2	19.16	11.24	0.805	20.75	0.07642	ok
Gangapur Road	NO2 *	469	47.71	6.907	3.419	-15.93	2.364	-1.922	ok
Gangapur Road	SO2	4034	6.064	2.462	1.294	0.4193	2.697	0.08691	ok
Gangapur Road	CO *	79	0.003984	0.06312	0.04723	-0.6252	0.02868	-1.2	unreliable (n < 168)
Gangapur Road	O3	4361	72.19	8.496	4.385	0.5914	7.437	-0.1424	ok
Gangapur Road	NH3 *	214	25.24	5.024	1.953	-584	0.1872	-25.84	ok
Pandav Nagari	PM2.5	0	-	-	-	-	-	-	undefined
Pandav Nagari	PM10	0	-	-	-	-	-	-	undefined
Pandav Nagari	NO2	0	-	-	-	-	-	-	undefined
Pandav Nagari	SO2	0	-	-	-	-	-	-	undefined
Pandav Nagari	CO	0	-	-	-	-	-	-	undefined
Pandav Nagari	O3	0	-	-	-	-	-	-	undefined
Pandav Nagari	NH3	0	-	-	-	-	-	-	undefined
Hirwadi	PM2.5	4629	22.31	4.724	3.213	0.9239	4.79	0.01381	ok
Hirwadi	PM10	4624	274.2	16.56	7.67	0.7891	18.45	0.1028	ok
Hirwadi	NO2	4254	15.45	3.93	2.42	0.6156	3.547	-0.108	ok
Hirwadi	SO2	4408	75.85	8.709	2.919	0.5011	8.723	0.001543	ok
Hirwadi	CO	1989	0.02544	0.1595	0.07564	0.7565	0.1508	-0.05772	ok
Hirwadi	O3	4638	161	12.69	6.185	0.7388	13.69	0.07313	ok
Hirwadi	NH3	4432	64.47	8.029	3.682	0.2151	8.397	0.04383	ok

Table 21. 2026: one-step predictions, unflagged stations pooled per pollutant.

Pollutant	n	RMSE	R^2	RMSE persist.	Skill vs persist.	Stations
PM2.5	13438	8.368	0.7982	8.436	0.008069	MIDC Ambad, Gangapur Road, Hirwadi
PM10	13435	17.07	0.8264	19.06	0.1042	MIDC Ambad, Gangapur Road, Hirwadi
NO2	8721	4.48	0.8205	4.932	0.09148	MIDC Ambad, Hirwadi
SO2	12984	5.431	0.6373	5.502	0.01283	MIDC Ambad, Gangapur Road, Hirwadi
CO	1989	0.1595	0.7565	0.1508	-0.05772	Hirwadi
O3	13551	9.634	0.7879	10.09	0.04489	MIDC Ambad, Gangapur Road, Hirwadi
NH3	8921	5.745	0.2845	5.999	0.04238	MIDC Ambad, Hirwadi

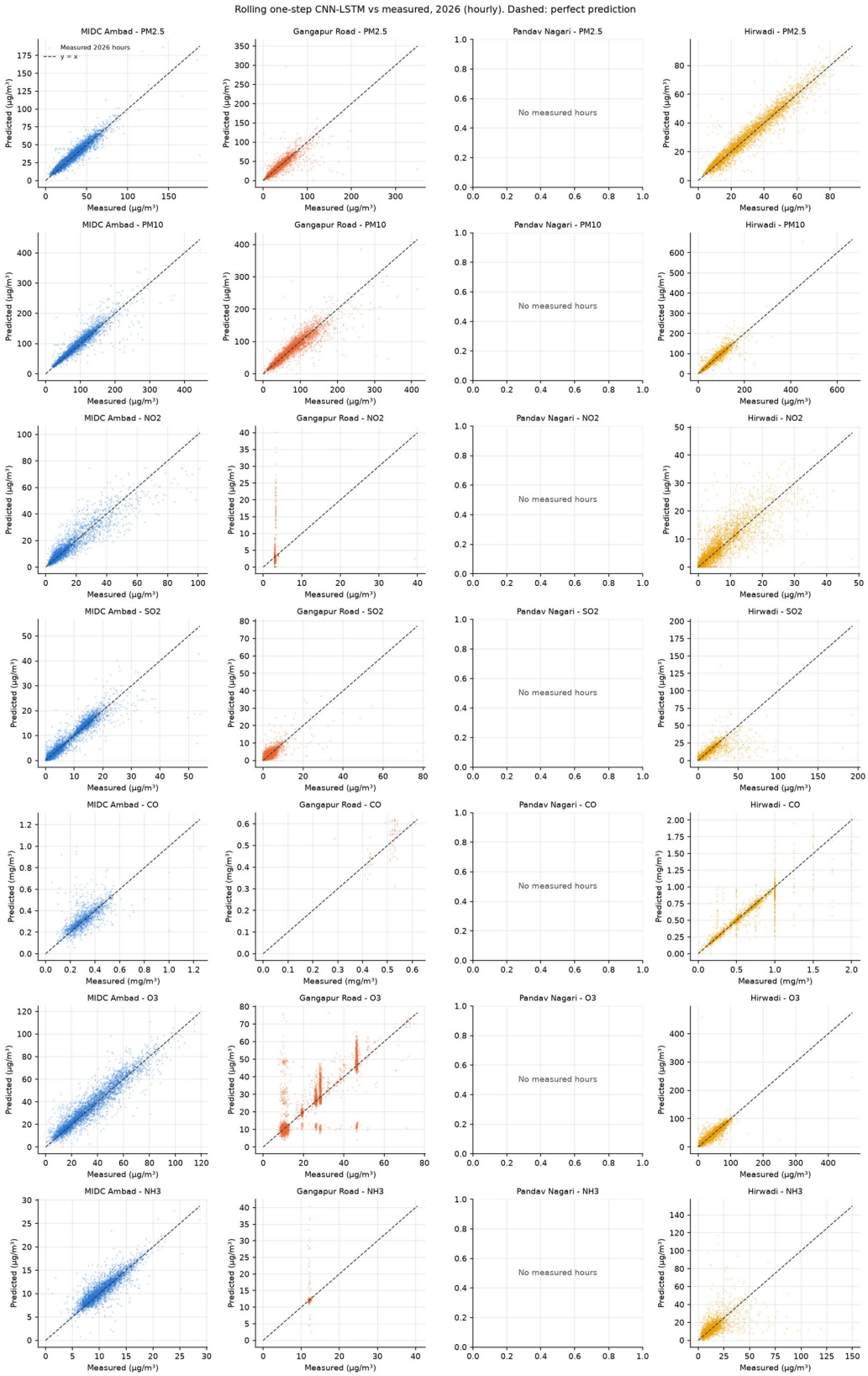


Figure 18. 2026 one-step predictions against measurements. Pandav Nagari has no 2026 measurements.

13. Predicted AQI

The model predicts pollutant concentrations; AQI is derived from them afterwards with the CPCB National AQI method [11]. Sub-indices use 24-hour means for PM2.5, PM10, NO2, SO2 and NH3 and the maximum 8-hour mean for CO and O3. The AQI is the largest sub-index and is computed only when at least three sub-indices, including PM2.5 or PM10, are available. A measured daily mean needs at least 16 measured hours. The Severe band has no published upper concentration, so it is extended linearly with the width of the Very Poor band and capped at 500. Predicted AQI is a model estimate from the recursive forecast. Measured AQI is computed from the station data with the same rules, not taken from an official bulletin.

Table 22. 2025 backtest: daily AQI from the recursive forecast against measured AQI.

Station	Forecast days	Days with measured AQI	Category agreement %	AQI RMSE	Mean predicted AQI	Mean measured AQI
MIDC Ambad	365	349	10.03	212.9	286.3	95.66
Gangapur Road	365	352	25.85	106.1	155.9	79.58
Pandav Nagari	365	347	48.41	47	60.81	79.27
Hirwadi	365	358	60.89	32.74	78.82	88.42

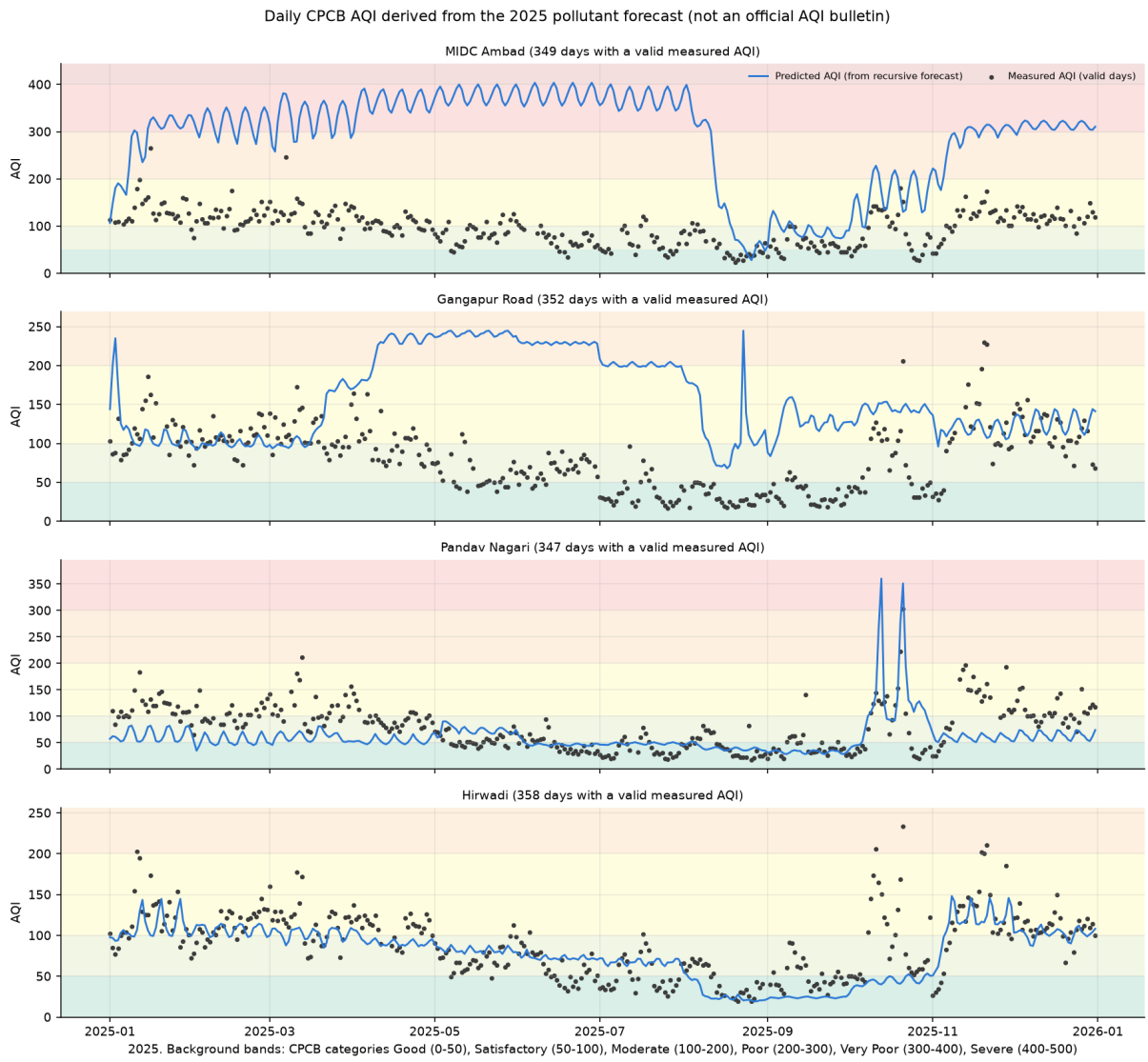


Figure 19. 2025 backtest: daily AQI derived from the forecast, with measured AQI on valid days. Bands: CPCB categories.

Table 23. 2026 forecast: daily AQI from the recursive forecast against measured AQI.

Station	Forecast days	Days with measured AQI	Category agreement %	AQI RMSE	Mean predicted AQI	Mean measured AQI
MIDC Ambad	365	196	53.57	25.95	90.43	88.26
Gangapur Road	365	206	35.44	31.73	96.24	76.01
Pandav Nagari	365	0	-	-	128.4	-
Hirwadi	365	198	1.515	148.6	199.2	72.98

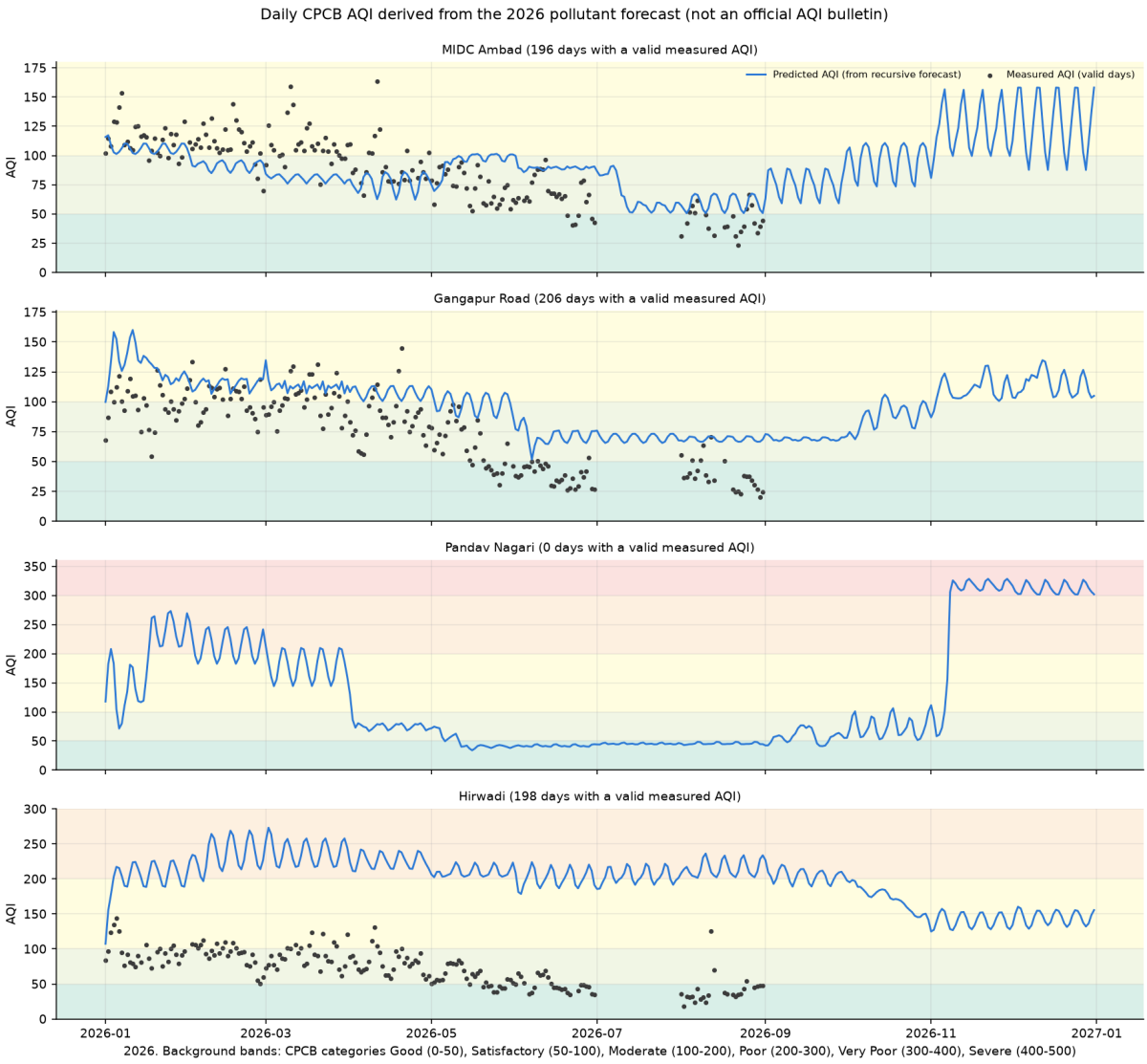


Figure 20. 2026 forecast: daily AQI derived from the forecast, with measured AQI on valid days. Bands: CPCB categories.

14. Discussion

Leakage checks. A separate script (scripts/verify_outputs.py) runs for each mode and re-derives the results from the raw CSV files and the saved models. It replays the first 48 hours of every station's recursive forecast from raw data truncated at the cutoff and reproduces the export, so the forecast does not depend on any later measurement. It rebuilds 20 random one-step input windows per station from the raw files, confirming that each prediction uses exactly the 24 preceding hours. It also checks that the scaler means equal the training-period means, that every horizon hour is present once, that missing hours carry no actual value, and that every metric matches a recomputation from the exported predictions.

Quality-flagged series, 2025 backtest: none after cleaning.

Quality-flagged series, 2026 forecast: MIDC Ambad CO, Gangapur Road NO2, Gangapur Road CO, Gangapur Road NH3 still take fewer than 100 distinct measured values after cleaning. They are reported in every table but excluded from pooled summaries and win counts.

Table 24. Horizon hours whose measurement was removed by the cleaning rules and is therefore not scored (out of 8,760 per series).

Mode	Station	PM2.5	PM10	NO2	SO2	CO	O3	NH3
2025 backtest	Gangapur Road	0	0	3057	0	469	575	3039
2025 backtest	Hirwadi	0	0	0	24	3	0	0
2025 backtest	MIDC Ambad	22	0	0	48	94	0	143
2025 backtest	Pandav Nagari	0	0	0	24	7	0	157
2026 forecast	Gangapur Road	0	0	4498	122	4901	16	4752
2026 forecast	Hirwadi	0	0	212	24	2672	0	84
2026 forecast	MIDC Ambad	0	0	0	0	3077	0	12
2026 forecast	Pandav Nagari	0	0	0	0	0	0	0

Recursive versus one-step. The two scores answer different questions. The one-step score says how well the network corrects persistence when fresh measurements arrive every hour. The recursive score says what can be said about a given hour months in advance, which is mostly its seasonal and diurnal expectation. The 2026 forecast file is the recursive forecast. The one-step results show what an operational next-hour system would achieve once 2026 data arrives.

15. Limitations

- 2026 ground truth covers January-June and August at three stations only; July and September-December are unmeasured, and Pandav Nagari has no 2026 data. The 2026 scores therefore miss most of the monsoon and the post-monsoon season.
- A recursive hourly forecast months ahead carries no information about specific weather events. Its hourly values should be read as an expected pattern, not as a prediction of individual hours.
- No uncertainty intervals: one training run per model (seed 42). Recursive errors accumulate with lead time, as the RMSE-by-month figure shows.
- Three stations have about two and a half years of training data (mid-2023 to the cutoff), and Gangapur Road's long history includes years whose sensor behaviour differs from 2026.
- The model uses no weather inputs (temperature, humidity, wind, rain), although they are measured at three of the four stations. Weather drives much of the day-to-day change in pollution, and a year-ahead forecast could only use it with weather forecasts, which this project does not have.
- The cleaning rules (12-hour stuck values, CO zeros, flatline days) and the distinct-value flag are heuristics; some degraded readings may remain, such as low-resolution CO in mid-2025.

- Derived AQI depends on the CPCB breakpoints as implemented here, and on the assumption for the open-ended Severe band.

16. Conclusion

The pipeline produces a leakage-free, timestamp-aligned hourly forecast of all of 2026 for all four stations, and scores it only where measurements exist. The independent verifier and the test suite confirm the horizon, the labelling of missing ground truth, the absence of post-cutoff information, and every metric.

Accuracy is a separate question, and the answer is not flattering. As a year-ahead forecast, the recursive CNN-LSTM is worse than the month-by-hour climatology for every pollutant, pooled, in the 2025 backtest (-223.0% (PM2.5) to -20.9% (PM10)) and for 6 of 7 pollutants in the measured part of 2026 (-211.8% (PM10) to 10.6% (CO)). Per series it beats climatology in only 3 of 28 (2025) and 2 of 17 (2026) cases. For planning-horizon estimates of 2026, the climatology column of the same forecast file is the better number. The network earns its place one hour ahead, modestly: with the previous 24 measured hours as input, it beats persistence for 22 of 28 series in 2025 and 12 of 17 in 2026 (median RMSE reduction 5.8% and 4.4%). Pooled per pollutant, its RMSE skill against persistence is -3.7% (NO2) to 8.1% (SO2) in 2025 and -5.8% (CO) to 10.4% (PM10) in 2026.

Future work: train a direct multi-horizon model, or blend the network with climatology using weights chosen on validation, rather than feeding a one-step model its own output for 8,760 steps; add quality-controlled meteorological inputs; report variance over several seeds; and re-score 2026 when July and September-December data and Pandav Nagari's 2026 data become available.

17. Reproducing the Results

```
NASHIK_MODE=backtest_2025 python nashik_aq_pipeline.py

NASHIK_MODE=forecast_2026 python nashik_aq_pipeline.py

python scripts/verify_outputs.py backtest_2025

python scripts/verify_outputs.py forecast_2026

python -m unittest discover tests -v

python scripts/build_report.py
```

Outputs are written to outputs/backtest_2025/ and outputs/forecast_2026/. run_metadata.json in each folder records the cutoff, horizon, features, target order, selected designs, model files, evaluation coverage, software versions and git commit (2ee0a80).

18. References

[1] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997, doi: 10.1162/neco.1997.9.8.1735.

[2] O. Troyanskaya et al., "Missing value estimation methods for DNA microarrays," Bioinformatics, vol. 17, no. 6, pp. 520-525, 2001, doi: 10.1093/bioinformatics/17.6.520.

[3] C.-J. Huang and P.-H. Kuo, "A Deep CNN-LSTM Model for Particulate Matter (PM2.5) Forecasting in Smart Cities," Sensors, vol. 18, no. 7, art. 2220, 2018, doi: 10.3390/s18072220.

[4] D. Qin et al., "A Novel Combined Prediction Scheme Based on CNN and LSTM for Urban PM2.5 Concentration," IEEE Access, vol. 7, pp. 20050-20059, 2019, doi: 10.1109/ACCESS.2019.2897028.

[5] H. Junninen et al., "Methods for imputation of missing values in air quality data sets," Atmospheric Environment, vol. 38, no. 18, pp. 2895-2907, 2004, doi: 10.1016/j.atmosenv.2004.02.026.

[6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

[7] Y. LeCun, L. Bottou, Y. Bengio and P. Haffner, "Gradient-based learning applied to document recognition," Proceedings of the IEEE, vol. 86, no. 11, pp. 2278-2324, 1998, doi: 10.1109/5.726791.

[8] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," arXiv:1412.6980, 2014 (ICLR 2015).

[9] N. Srivastava et al., "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," Journal of Machine Learning Research, vol. 15, pp. 1929-1958, 2014.

[10] C. Bergmeir and J. M. Benítez, "On the use of cross-validation for time series predictor evaluation," Information Sciences, vol. 191, pp. 192-213, 2012, doi: 10.1016/j.ins.2011.12.028.

[11] Central Pollution Control Board, "National Air Quality Index," Ministry of Environment, Forest and Climate Change, Government of India, 2014.